Tail Rotor Dynamics for an RC Helicopter

Thrust, Induced Inflow, and Inner-Loop Inversion

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Abstract

This note develops a control-oriented model of an RC helicopter tail rotor. Starting from blade-element momentum theory (BEMT), we derive the thrust-versus-pitch relation, add dynamic inflow via the Pitt–Peters first-order model, and invert the coupled system to obtain the blade pitch required to track a commanded thrust. The inverse is collected into a compact implementation-ready form for the inner control loop. A linearisation about hover trim is also developed, exposing the lead–lag structure of the plant and quantifying the inflow-softening factor that varies across the operating envelope. The axial flow through the disc is shown to include a body-yaw-rate contribution $\ell_t\omega$ that is of order the hover induced velocity at RC 3D yaw rates and must be carried explicitly in the inverse. Realistic parameter ranges for RC tail rotors are tabulated.

Contents

Nomenclature

The notation used throughout the note is collected below. Overbars denote trim (equilibrium) values; $\delta(\cdot)$ denotes small perturbations about trim; a superscript "cmd" denotes a commanded (reference) quantity. SI units are used unless otherwise stated; angles are in radians except where a table lists degrees.

Rotor geometry and operating condition

Symbol	Description	Units
R_t	Tail-rotor radius	m
r	Radial coordinate of a blade element, $0 \le r \le R_t$	m
c	Blade chord (assumed rectangular blade)	m
N_b	Number of blades	–
A_t	Rotor-disc area, $A_t = \pi R_t^2$	m^2
σ	Rotor solidity, $\sigma = N_b c/(\pi R_t)$	–
Ω_t	Rotor angular speed	$\mathrm{rad\,s^{-1}}$
$\Omega_t R_t$	Blade tip speed	$\mathrm{m\,s^{-1}}$
ρ	Air density (taken as $1.225\,\mathrm{kg\,m^{-3}}$)	$\mathrm{kg\,m^{-3}}$
Re	Blade-section Reynolds number	–

Blade-element aerodynamics

Symbol	Description	Units
θ_t	Blade pitch angle (control input)	rad
U_T	Tangential component of local flow at a blade section	$\mathrm{m\,s^{-1}}$
U_P	Perpendicular (axial) component of local flow	$\mathrm{m\,s^{-1}}$
ϕ	Local inflow angle, $\phi \approx U_P/U_T$	rad
α	Effective angle of attack, $\alpha = \theta_t - \phi$	rad
a	Two-dimensional lift-curve slope ($\approx 5.7\,\mathrm{rad^{-1}}$)	$\mathrm{rad^{-1}}$
C_ℓ	Section lift coefficient, $C_\ell = a\,\alpha$	–

Inflow and free-stream velocities

Symbol	Description	Units
v_i	Uniform induced velocity through the rotor disc (state)	$\mathrm{m\,s^{-1}}$
$v_{i,\mathrm{ss}}$	Steady-state value of v_i for the current operating point	$\mathrm{m\,s^{-1}}$
v_h	Hover induced velocity, $v_h = \sqrt{T_t/(2\rho A_t)}$	$\mathrm{m\,s^{-1}}$
V_∞	Axial flow through the rotor disc, $V_\infty = V_\infty^{\mathrm{aero}} + \ell_t \omega$	$\mathrm{m\,s^{-1}}$
V_∞^{aero}	Airframe-translation contribution to axial flow	$\mathrm{m\,s^{-1}}$
$V_\perp$	In-plane component of the free-stream velocity	$\mathrm{m\,s^{-1}}$
ω	Body yaw rate	$\mathrm{rad\,s^{-1}}$
ℓ_t	Moment arm from airframe CG to tail-rotor hub	m
w_∞	Axial velocity in the far wake	$\mathrm{m\,s^{-1}}$
U_{disc}	Resultant velocity through/sweeping the actuator disc	$\mathrm{m\,s^{-1}}$
μ_z	Non-dimensional axial inflow, $\mu_z = V_\infty/(\Omega_t R_t)$	–
μ	Advance ratio (in-plane), $\mu = V_\perp/(\Omega_t R_t)$	–

Thrust and non-dimensional coefficients

Symbol	Description	Units
T_t	Tail-rotor thrust	N
$\dot{m}$	Mass-flow rate through the actuator disc	$\mathrm{kg\,s^{-1}}$
Δp	Pressure jump across the actuator disc	Pa
C_T	Thrust coefficient, $C_T = T_t/[\rho A_t(\Omega_t R_t)^2]$	–
λ	Total inflow ratio, $\lambda = (v_i + V_\infty)/(\Omega_t R_t)$	–
λ_i	Induced-inflow ratio, $\lambda_i = v_i/(\Omega_t R_t)$	–
λ_h	Hover inflow ratio, $\lambda_h = \sqrt{C_T/2}$	–

Dynamic inflow model

Symbol	Description	Units
τ_i	First-order inflow time constant (Pitt–Peters)	s
0.849	Apparent-mass coefficient, $8/(3\pi)$, in τ_i	–

Trim (equilibrium) quantities

Symbol	Description	Units
$\bar{T}$	Trim thrust (typically hover)	N
$\bar{v}_i$	Trim induced velocity	$\mathrm{m\,s^{-1}}$
$\bar{\theta}$	Trim blade pitch	rad
$\bar{\lambda}$	Trim inflow ratio	–
$\bar{C}_T$	Trim thrust coefficient	–
$\bar{\tau}_i$	Trim inflow time constant	s

Linearisation, plant model and compensator

Symbol	Description	Units
$\delta(\cdot)$	Small perturbation of a quantity about its trim value	(varies)
s	Laplace variable	$\mathrm{s^{-1}}$
t	Time	s
G_0	Frozen-wake thrust-per-radian gain, eq. (43)	$\mathrm{N\,rad^{-1}}$
k_v	Linearised sensitivity $\partial v_{i,\mathrm{ss}}/\partial T$, eq. (44)	$\mathrm{m\,s^{-1}\,N^{-1}}$
β	Dimensionless inflow-thrust coupling, $\beta = \sigma a/(16\lambda_h)$, eq. (50)	–
$S(\lambda)$	Self-consistent BEMT softening factor $1/(1+\beta)$, eq. (61)	–
x	Internal state of the lead compensator	rad

Non-dimensional implementation form (Section 6)

Symbol	Description	Units
T_{ref}	Reference thrust ($u = 1$ maps to this thrust)	N
v_{ref}	Reference velocity, $\sqrt{T_{\mathrm{ref}}/(2\rho A_t)}$, eq. (63)	$\mathrm{m\,s^{-1}}$
λ_{ref}	Reference inflow ratio, $v_{\mathrm{ref}}/(\Omega_t R_t)$	–
τ_{ref}	Reference wake timescale, $0.849\,R_t/(4v_{\mathrm{ref}})$	s
u	Dimensionless thrust command, $T_t^{\mathrm{cmd}}/T_{\mathrm{ref}}$	–
$\hat{v}_i, \hat{v}_{i,\mathrm{ss}}$	Dimensionless induced velocity (state, steady target)	–
$\hat{V}_\infty, \hat{V}_\perp$	Dimensionless axial and edgewise free-stream velocities	–
K_u	Pitch-per-command coefficient, eq. (69)	rad
K_v	Pitch-per-through-disc-velocity coefficient, eq. (70)	rad
K_τ	Discrete inflow-update coefficient, eq. (75)	–
K_ω	Yaw-rate to dimensionless axial-flow scale, eq. (76)	s

Commanded (reference) quantities

Symbol	Description	Units
T_t^{cmd}	Commanded tail-rotor thrust	N

Acronyms

BEMT	Blade Element Momentum Theory
DC	Direct current (steady-state, low-frequency limit)
TF	Transfer function
RC	Radio-controlled

1 Introduction

The tail rotor of a single-main-rotor helicopter is, in steady flight, a near-quasi-steady thrust generator: its job is to balance the anti-torque of the main rotor and, via the pilot's pedal input, to provide directional control. On a radio-controlled (RC) model helicopter the same physics applies, but the rotor operates at much lower Reynolds number, much higher rotational speed, and with a single pitch input. From a flight-control perspective, the relevant question is how the commanded blade pitch θ_t maps to the thrust T_t actually produced, in both steady and transient conditions.

Two effects make this map non-trivial. First, the thrust produced at a given pitch depends on the induced velocity v_i through the rotor: as the rotor generates thrust, it accelerates air through the disc, which changes the local angle of attack of every blade element and reduces the effective incidence. Second, the induced velocity itself takes a finite time to develop, because the wake has inertia; the relationship between v_i and T_t is therefore dynamic, not algebraic. The combined effect is a *lead–lag* response in thrust to a step in pitch: an immediate transient at the open-loop (no-inflow) gain, followed by partial relaxation to a lower steady value as inflow builds up.

This note develops a compact, control-oriented model of these effects. Section 2 derives the blade-element thrust relation, an expression linear in pitch and in induced velocity. Section 3 closes the loop on v_i using momentum theory and the first-order Pitt–Peters dynamic-inflow model. Section 4 inverts the coupled system to obtain the pitch required to track a thrust command. Section 5 develops the transfer function about hover trim, exposing its lead–lag structure and quantifying the *inflow-softening factor* that varies across the operating envelope. Section 6 recasts the inverse in dimensionless form, showing that every rotor and air parameter collapses into four dimensionless coefficients and quantifying how those coefficients scale with helicopter size. Section 7 collects the inverse into a compact form suitable for direct implementation in an inner control loop. Section 8 tabulates parameter ranges for typical RC-helicopter classes and works a representative numerical example. Section 9 steps outside the tail rotor itself to derive the airframe yaw equation that the inner loop drives, showing how the main rotor's drag, the tail rotor's thrust, and the inertias of the airframe and the main rotor combine into a single first-order equation for the yaw rate ω.

2 Blade Pitch and Rotor Thrust

Section 1 set out the central problem: predict the thrust T_t that a tail rotor produces given the pitch θ_t applied to its blades. This section builds the first half of that prediction—an expression for thrust from the geometry and kinematics of a single rotating blade, at a frozen instant in time, conditional on the induced velocity v_i through the disc. The second half (computing v_i itself) is taken up in Section 3.

The tool is *Blade Element Theory* (BET), which decomposes the rotor into thin radial strips, treats each strip as a two-dimensional aerofoil section, and integrates the resulting elemental lift across span and around the disc. Combined with momentum theory in Section 3 it becomes *Blade Element Momentum Theory* (BEMT). The blade-element half is purely kinematic and aerodynamic; it does not invoke any global mass or momentum balance. That is what makes it tractable as a standalone derivation.

2.1 The Rotor and the Blade-Element Abstraction

We model the tail rotor as N_b identical blades of length R_t and chord c, rotating at constant angular speed Ω_t about a shaft, sweeping out a disc of area $A_t = \pi R_t^2$. For an RC tail rotor N_b is typically 2; for full-scale tail rotors it may be 3 or 4. The blades are set to a single pitch θ_t common to all blades and to every azimuthal position. This is a meaningful simplification specific to tail rotors: main rotors additionally vary their pitch sinusoidally with azimuth (cyclic pitch) to control the disc's tilt, but tail rotors normally do not—they have only one degree of freedom, the blade pitch θ_t, which acts as the thrust command. (We avoid the aerodynamic term *collective* throughout: in RC helicopter usage "collective" refers exclusively to the main rotor's pitch input, so this document uses "pitch" or "blade pitch" for the tail.)

The blade-element idea is to chop each blade into thin radial strips of width dr at radial station r, treat each strip as an isolated two-dimensional aerofoil section that sees its own local apparent wind, compute the elemental lift and drag on the strip from 2D aerofoil theory, then sum (integrate) over the span and multiply by N_b. The abstraction works because the spanwise extent of a strip is large compared with the chord: lift on any given strip is dominated by 2D mechanisms and depends only weakly on the flow at neighbouring spanwise stations. The cost is that 3D effects—tip vortices, trailing-edge wake roll-up, root flow distortion—are excluded by construction and must be patched back in afterwards if needed (see Section 2.7).

2.2 Velocity Components at the Blade Element

At radial station r on the blade, two velocity components of the apparent wind matter. The first is *tangential*, lying in the plane of the disc and pointing opposite to the blade's motion:

$$U_T \;=\; \Omega_t\, r, \tag{1}$$

zero at the hub, linear in r, peaking at $\Omega_t R_t$ at the tip. This is what the blade sees just from rotating: the element is moving forward through still air at speed $\Omega_t r$, so the air appears to move backward past it at the same speed.

The second is *perpendicular*, normal to the disc plane, pointing through the disc:

$$U_P \;=\; v_i \;+\; V_\infty. \tag{2}$$

v_i is the induced velocity produced by the rotor itself (present whenever the rotor produces thrust, drawn through the disc by the pressure jump the blades create), and V_∞ is any external axial component of the freestream—the air flow aligned with the rotor shaft. Two physical sources contribute to it:

$$V_\infty \;=\; V_\infty^{\text{aero}} \;+\; \ell_t\, \omega. \tag{3}$$

The first term, V_∞^{aero}, is the airframe-translation contribution: for a tail rotor on a sideways-translating airframe, this is the sideways translation along the rotor shaft. The second term is the velocity of the tail-rotor hub through the air due to body yaw rate ω about the airframe CG, with moment arm ℓ_t; the hub moves in an arc whose tangent at the rotor is along the rotor shaft, so this velocity is purely axial. The sign convention follows T_t: $\omega > 0$ when the airframe rotates in the same sense the tail rotor's thrust is pushing it, so a sustained yaw in the "with-thrust" direction produces $V_\infty > 0$—air fed into the suction side of the disc, partially unloading the rotor. The yaw-rate contribution is quantified in Section 3.3 and shown there to be a first-order effect at the yaw rates encountered in RC operation, not a small correction. In hover with no airframe motion, both contributions vanish and the perpendicular flow is purely the rotor's own induced velocity. The two notations V_∞ and $\mu_z \Omega_t R_t$ refer to the same quantity, with μ_z the non-dimensional form used in some textbooks.

The velocity triangle. Imagine looking edge-on at the blade element from outboard, so its chord is perpendicular to the page. The element moves forward in the disc plane at speed U_T, and at the same time air is being drawn through the disc at speed U_P. In the element's frame these combine into an apparent wind of magnitude $U = \sqrt{U_T^2 + U_P^2}$ blowing toward the element at an angle

$$\phi \;=\; \tan^{-1}\!\left(\frac{U_P}{U_T}\right) \;\approx\; \frac{U_P}{U_T} \tag{4}$$

out of the disc plane. ϕ is the *inflow angle*: the amount by which the apparent wind is tilted out of the plane in which the blade moves. Away from the root—where U_T is small and the small-angle approximation fails—typical values are $U_P \sim$ a few m/s against $U_T \sim$ tens of m/s, so $\phi \sim 0.05$–$0.15\,\text{rad}$. Small-angle is justified across most of the span.

2.3 Effective Angle of Attack and Elemental Thrust

The blade is set at geometric pitch θ_t relative to the disc plane. But the section aerofoil cares only about its angle to the *apparent wind*, not to the disc. Since the apparent wind is tilted by ϕ out of the disc plane, the angle that the chord line makes with the flow—the *effective angle of attack*—is reduced:

$$\alpha \;=\; \theta_t \;-\; \phi. \tag{5}$$

This is the central kinematic mechanism of this section. The geometric pitch θ_t is what we command via the servo, but the element only feels α, and induced inflow—through ϕ—steals incidence from the blade. Every newton of thrust the rotor produces makes its own future thrust slightly harder to produce. This is what the integrated result will quantify, and what the rest of the note must contend with.

From angle of attack to elemental lift. Assume a linear two-dimensional lift curve, $C_\ell = a\,\alpha$, with slope a. For an inviscid thin symmetric aerofoil the slope is 2π per radian ($\approx 6.28\,\text{rad}^{-1}$); real aerofoils at typical RC tail rotor Reynolds numbers give $a \approx 5.5$–$5.9\,\text{rad}^{-1}$, and we adopt $a \approx 5.7\,\text{rad}^{-1}$ as a representative value. The elemental lift on a strip of width dr at station r is then

$$dL \;=\; \tfrac{1}{2}\rho U^2\, c\, C_\ell\, dr \;=\; \tfrac{1}{2}\rho\,(U_T^2 + U_P^2)\, c\, a\, \alpha\, dr, \tag{6}$$

acting perpendicular to the apparent wind.

From elemental lift to elemental thrust. Thrust on the rotor is the component of aerodynamic force along the rotor axis, i.e. perpendicular to the disc plane. The lift vector sits perpendicular to the apparent wind, which is itself tilted by ϕ out of the disc plane, so the lift is tilted by ϕ *away from* the rotor axis. There is also a drag vector aligned with the apparent wind, contributing in the opposite sense. The exact projection is

$$dT \;=\; dL\cos\phi \;-\; dD\sin\phi. \tag{7}$$

For small ϕ, $\cos\phi \approx 1$ (the lift projects almost entirely onto the rotor axis), and $\sin\phi \approx \phi$ so the drag term is of order $\phi\, C_d/C_\ell$ times the lift term—small, since ϕ is small and C_d is much less than C_ℓ at typical angles of attack. Dropping both corrections, and additionally neglecting U_P^2 against U_T^2 in the apparent-wind magnitude (consistent with ϕ small), we obtain

$$dT \;\approx\; \tfrac{1}{2}\rho\,(\Omega_t r)^2\, c\, a\,(\theta_t - \phi)\, dr. \tag{8}$$

Each small-angle simplification is itself $O(\phi^2)$; collectively they limit the validity of the derivation to a band of mostly small inflow angles.

2.4 Integration Across the Span

Total rotor thrust is the elemental thrust (8) summed across one blade's span and across all N_b blades:

$$T_t \;=\; N_b \int_0^{R_t} dT \;=\; \tfrac{1}{2}\rho\, a\, c\, N_b\, \Omega_t^2 \int_0^{R_t} (\theta_t - \phi)\, r^2\, dr. \tag{9}$$

Assume the induced velocity is *uniform* across the disc—i.e. v_i does not vary with r or with azimuth. This is the standard BEMT simplification. Under it, U_P is constant in r, and $\phi(r) = U_P/(\Omega_t r) = (v_i + V_\infty)/(\Omega_t r)$ varies with r only through its denominator. The integrand splits into two pieces:

$$\int_0^{R_t} \theta_t\, r^2\, dr \;=\; \frac{\theta_t\, R_t^3}{3}, \qquad \int_0^{R_t} \phi(r)\, r^2\, dr \;=\; \frac{v_i + V_\infty}{\Omega_t}\,\frac{R_t^2}{2}. \tag{10}$$

Both pieces are tip-weighted—the pitch contribution as r^2 and the inflow loss as r—because elemental thrust scales with dynamic pressure, which goes as $(\Omega_t r)^2$. Outboard stations do most of the work; inboard stations are nearly irrelevant. This is the reason rotors are conventionally analysed in terms of tip quantities.

2.5 Non-Dimensional Form

Substituting the two integrals back into the thrust expression and collecting:

$$T_t \;=\; \tfrac{1}{2}\rho\, a\, c\, N_b\, \Omega_t^2 \left[\frac{\theta_t\, R_t^3}{3} - \frac{v_i + V_\infty}{\Omega_t}\,\frac{R_t^2}{2}\right]. \tag{11}$$

Divide both sides by $\rho\, A_t\, (\Omega_t R_t)^2$, the natural dynamic scale (air density times disc area times tip-speed squared). Two non-dimensional groups emerge:

$$\sigma \;\equiv\; \frac{N_b\, c}{\pi R_t}, \qquad \lambda \;\equiv\; \frac{v_i + V_\infty}{\Omega_t R_t}. \tag{12}$$

σ is the *rotor solidity*: the fraction of the disc's swept area that is actually covered by blade planform. It is "how much rotor there is per unit disc area." Typical RC tail rotors have $\sigma \approx 0.10$–0.20, somewhat higher than main rotors ($\sigma \approx 0.05$–0.10) because tail rotors are short-radius and need proportionally more blade chord to generate their share of thrust. λ is the *inflow ratio*: the through-disc velocity expressed as a fraction of tip speed. It is small (~ 0.05–0.15 in hover) because the tip moves much faster than the wake.

The thrust coefficient

$$C_T \;\equiv\; \frac{T_t}{\rho\, A_t\, (\Omega_t R_t)^2} \tag{13}$$

is thrust non-dimensionalised by the same scale—a kind of "thrust per unit tip dynamic pressure per unit disc area." Substituting these groups gives the standard BEMT result:

$$\boxed{C_T \;=\; \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right).} \tag{14}$$

The factor $1/3$ on θ_t is the r^2 weighting of the pitch contribution; the factor $1/2$ on λ is the r weighting of the inflow loss. Dimensional thrust is recovered as

$$T_t \;=\; C_T\,\rho\,A_t\,(\Omega_t R_t)^2. \tag{15}$$

2.6 Dimensional Form

Equation (14) expresses thrust through the dimensionless coefficient C_T. For the rest of this note it is more convenient to carry thrust in newtons directly. Multiplying through by the natural dimensional scale $\rho A_t(\Omega_t R_t)^2$, and writing the inflow ratio in expanded form $\lambda = (v_i + V_\infty)/(\Omega_t R_t)$:

$$\boxed{T_t \;=\; \rho A_t(\Omega_t R_t)^2\,\frac{\sigma a}{2}\left(\frac{\theta_t}{3} \;-\; \frac{v_i + V_\infty}{2\,\Omega_t R_t}\right).} \tag{16}$$

At fixed Ω_t (held by the main-rotor governor on an RC helicopter), the prefactor and the aerodynamic group σa are constants of the rotor: they depend on geometry (R_t, c, N_b), air density, lift-curve slope, and tip speed, but not on the operating point. *Equation (16) is exact within the BEMT assumptions listed below*; it is not a linearisation about any particular point—the linearity in θ_t, v_i, and V_∞ is a property of the underlying derivation.

Reading the two terms. The first term, proportional to $\theta_t/3$, is the thrust the rotor would produce per radian of blade pitch if no induced inflow developed: the open-loop or "frozen-wake" pitch effectiveness, scaling as $(\Omega_t R_t)^2$ (it is dynamic-pressure-like). The second, proportional to $(v_i + V_\infty)/(2\Omega_t R_t)$, is the rate at which thrust is given back as inflow builds: each m/s of through-disc flow costs a proportional amount of thrust, scaling only as $\Omega_t R_t$ (it is velocity-like). The ratio of inflow penalty to pitch effectiveness is therefore purely kinematic, set by the factor

$$\frac{3}{2\,\Omega_t R_t}, \tag{17}$$

which is independent of blade geometry, density, lift-curve slope, or solidity, and which reappears as a structural feature in the inverse (Section 4) and in the dimensionless pitch law of Section 6. Faster-spinning rotors are less penalised by inflow, in the precise sense that (17) is smaller. The rest of this note is, in one sense, a careful study of how the second term in (16) evolves dynamically.

2.7 Assumptions and Where Each Is Repaired

Equation (16), and the BEMT result (14) from which it descends, rest on the following simplifications:

- *Uniform induced velocity v_i across the disc.* Repaired in Section 3 via momentum theory, which *supplies* the value of v_i given the thrust. The one-state Pitt–Peters extension makes v_i dynamic.
- *Small inflow angle,* $\phi \ll 1$, justifying $\tan\phi \approx \phi$, $\cos\phi \approx 1$, and $U_P^2 \ll U_T^2$. Holds across most of the span but breaks down near the root, where U_T is small; root cutout mitigates this.
- *Linear two-dimensional lift curve,* $C_\ell = a\alpha$, up to stall. Stall, Reynolds-number effects, and Mach-number effects (less relevant for RC) require empirical or table-lookup corrections to a.

- *Rectangular, untwisted blade with constant chord c.* Non-rectangular or twisted blades introduce r-dependent factors inside the span integrals but no change to the basic structure.
- *No tip loss and no root cutout* (integration from 0 to R_t). Tip loss is conventionally handled by multiplying C_T by a Prandtl tip-loss factor $B \approx 0.97$; root cutout by truncating the lower integration limit.
- *Thrust contribution from section drag is ignored* ($O(\phi^2)$, see Section 2.3).
- *Constant rotor speed Ω_t*, held by the governor.

Section 8 quantifies where each of these breaks down for RC-scale tail rotors. The first item—uniform inflow—is the load-bearing one for what follows: the rest of the document is, in a sense, a careful study of what happens once v_i is allowed its own dynamics.

3 Induced Inflow: Momentum Theory

The blade-element derivation of Section 2 produced $C_T = (\sigma a/2)(\theta_t/3 - \lambda/2)$, which contains *two* unknowns: the pitch θ_t (which we set) and the inflow ratio λ (which we do not). To close the system we need a second relationship between thrust and induced velocity. *Momentum theory* supplies it by treating the rotor disc as a whole and applying conservation of mass and momentum to the column of air passing through it.

For this section take positive velocity to mean flow in the thrust-producing direction through the rotor. Thus V_∞ is the already-existing axial flow through the disc, v_i is the extra velocity induced by the rotor, and $V_\infty + v_i$ is the through-disc velocity seen by the actuator disc. The theory below describes the normal working state of the rotor, with a smooth wake; strongly reversed flow, vortex-ring operation, and separated blade aerodynamics are outside this ideal model.

3.1 The Actuator Disc and the Mass-Flow Argument

Replace the physical rotor by an idealised *actuator disc*: an infinitesimally thin permeable surface of area A_t that imparts a uniform pressure jump Δp to the air passing through it. The disc does no modelling of individual blades; it is only a bookkeeping surface that adds energy to the flow. In the ideal actuator-disc model there is no swirl, no profile loss, and no non-uniform loading.

Draw a streamtube enclosing the disc. Station 1 is far upstream, where the pressure is ambient and the axial velocity is V_∞. Stations 2 and 3 are immediately on the upstream and downstream faces of the disc; the velocity is continuous through the disc, so both sides have axial velocity $V_\infty + v_i$, but the pressure jumps by Δp. Station 4 is far downstream, where the pressure has returned to ambient and the axial velocity is w_∞. In hover $V_\infty = 0$, so the upstream flow starts from rest and the streamtube contracts as the rotor pulls air through it.

Three relationships govern this column of air:

1. *Mass conservation.* The mass flow rate $\dot{m} = \rho A(x) U(x)$ is the same at every cross-section of the streamtube. The area $A(x)$ changes as the streamtube contracts, so only at the disc can we write the mass flow as
$$\dot{m} \;=\; \rho A_t\,(V_\infty + v_i), \tag{18}$$
where v_i is the induced velocity *at the disc.*

2. *Momentum.* The thrust on the rotor is the reaction to the change in axial momentum of the air, evaluated between far upstream and far downstream: $T_t = \dot{m}\,(w_\infty - V_\infty)$, where w_∞ is the axial velocity in the far wake.

3. *Bernoulli, separately on each side of the disc.* Energy is conserved along streamlines on each side of the disc, but *not* across it, because the rotor does work on the fluid. Applying Bernoulli from station 1 to station 2 and from station 3 to station 4 gives
$$p_\infty + \tfrac{1}{2}\rho V_\infty^2 = p_2 + \tfrac{1}{2}\rho(V_\infty + v_i)^2, \tag{19}$$
$$p_3 + \tfrac{1}{2}\rho(V_\infty + v_i)^2 = p_\infty + \tfrac{1}{2}\rho w_\infty^2. \tag{20}$$
The pressure jump is $\Delta p = p_3 - p_2$, and the disc force is $T_t = \Delta p\, A_t$. Subtracting the two Bernoulli equations gives
$$\frac{T_t}{A_t} \;=\; \frac{1}{2}\rho\,(w_\infty^2 - V_\infty^2). \tag{21}$$
Combining (21) with the momentum equation and (18) gives the celebrated *Froude result*:
$$w_\infty \;=\; V_\infty + 2\,v_i. \tag{22}$$

The Froude result is non-obvious and worth dwelling on: *half* the total velocity increment $w_\infty - V_\infty$ has already appeared at the disc, and the other half is added downstream. In hover this means the far wake moves at $2v_i$; by mass conservation its area is then $A_t/2$. This factor of two appears in every momentum-theory thrust expression and is the reason rotors are *efficient* thrust generators at low disc loading: they create thrust by moving a large mass of air by a moderate speed, rather than a small mass of air by a very high speed.

Substituting $w_\infty - V_\infty = 2v_i$ into the momentum equation, together with $\dot{m} = \rho A_t(V_\infty + v_i)$, gives the *axial-flight* form of the momentum balance:

$$T_t \;=\; 2\,\rho\,A_t\,v_i\,(V_\infty + v_i). \tag{23}$$

3.2 Edgewise Flow: The Full Momentum Balance

For a tail rotor mounted on a translating airframe, the free-stream is not generally aligned with the rotor axis. Decompose the airframe velocity at the rotor into a component along the rotor axis (axial, V_∞, which feeds the streamtube directly) and a component in the disc plane (edgewise, $V_\perp$, which sweeps air sideways through the disc). A useful way to picture this is as a tilted streamtube: the wake is still accelerated mainly in the rotor-axis direction, but fresh air is also swept through the disc from the side. That extra sweeping increases the mass flow through the actuator disc, so for a given thrust the required induced velocity is smaller than it would be in pure hover.

Glauert's approximation captures that effect by replacing the axial disc velocity in the mass-flow estimate with the resultant speed at the disc: the vector sum of the in-plane sweep $V_\perp$ and the through-disc flow $V_\infty + v_i$:

$$U_{\text{disc}} \;=\; \sqrt{(V_\infty + v_i)^2 + V_\perp^2}. \tag{24}$$

The mass-flow rate through the disc is then modelled as $\dot{m} = \rho A_t\, U_{\text{disc}}$, and the axial momentum imparted to the flow remains $2v_i$, as in the purely axial case. This is not a new exact Bernoulli derivation; it is the standard control-oriented momentum closure used for oblique inflow. The momentum equation becomes

$$\boxed{T_t \;=\; 2\,\rho\,A_t\,v_i\,\sqrt{(V_\infty + v_i)^2 + V_\perp^2}.} \tag{25}$$

The square-root factor is just the resultant velocity at the disc; the factor of two is Froude. In the purely axial case ($V_\perp = 0$) we recover (23), and in the hover limit ($V_\infty = V_\perp = 0$) the relation simplifies further to

$$T_t \;=\; 2\,\rho\,A_t\,v_i^2 \qquad \Longrightarrow \qquad v_h \;\equiv\; \sqrt{\frac{T_t}{2\,\rho\,A_t}}. \tag{26}$$

Reading the hover result. Equation (26) is the single most-cited result of rotor aerodynamics. It says induced velocity is set by *disc loading* T_t/A_t: rotors with larger disc area need less induced velocity for the same thrust, and induced power $T_t\,v_h \propto T_t^{3/2}/\sqrt{A_t}$ is lower as a result. A tail rotor (small A_t, high T_t/A_t) therefore has *high* v_i compared with a main rotor of equal thrust—a numerical example in Section 8 gives $v_h \sim 9\,\mathrm{m\,s^{-1}}$ for a typical 700 class tail rotor producing 12 N of thrust.

3.3 Yaw Rate as a Source of Axial Flow

The free-stream axial velocity V_∞ introduced in Section 2.2 has two sources, written together in (3) as $V_\infty = V_\infty^{\text{aero}} + \ell_t\omega$. The first is straightforward airframe translation along the rotor shaft. The second deserves a closer look because it is specific to yawing flight and is large enough at RC yaw rates to dominate the axial-flow contribution.

When the airframe yaws at rate ω about a vertical axis through its centre of mass, the tail rotor hub—located at distance ℓ_t behind the CG—swings through an arc at tangential velocity $\ell_t\,\omega$. Because the tail boom is normally perpendicular to the rotor shaft, the tangent to that arc at the rotor hub is aligned with the shaft. The yaw-induced hub motion is therefore axial through the disc, not edgewise, and it enters the momentum balance through V_∞ rather than $V_\perp$:

$$V_\infty \;=\; V_\infty^{\text{aero}} \;+\; \ell_t\,\omega, \qquad V_\perp \;=\; V_\perp^{\text{aero}} \quad \text{(no yaw-rate contribution).} \tag{27}$$

The sign convention is fixed by demanding self-consistency with the thrust direction. In the document's convention $V_\infty > 0$ means flow into the rotor in the same direction as the wake it produces, so $\omega > 0$ corresponds to airframe rotation in the same sense the tail-rotor thrust is pushing it—a "with-thrust" pirouette. In that case the rotor is moving *into* its own wake direction; air is fed into the suction side of the disc; and the rotor needs less of its own induced velocity to deliver a given thrust. An "against-thrust" pirouette gives $\omega < 0$ and pushes the operating point toward the windmill-brake or vortex-ring states where the smooth-wake momentum closure breaks down (Section 3.1).

Magnitude. RC helicopters routinely reach yaw rates of $\omega = 500\,^\circ\,\mathrm{s}^{-1} \approx 8.7\,\mathrm{rad\,s}^{-1}$ in 3D flight, with extreme pirouette flips going higher. For a 700 class machine with $\ell_t \approx 0.95\,\mathrm{m}$ the resulting axial velocity is $\ell_t\omega \approx 8.3\,\mathrm{m\,s}^{-1}$, comparable to the hover induced velocity $\bar{v}_i \approx 9\,\mathrm{m\,s}^{-1}$ of the same rotor (see Section 8.4). Smaller classes get smaller ℓ_t but proportionally smaller $\bar{v}_i$ too, so the dimensionless ratio $\ell_t\omega/\bar{v}_i$ reaches order unity across the size range. The non-dimensional axial inflow $\mu_z = \ell_t\omega/(\Omega_t R_t)$ is then of order $\lambda_h \approx 0.06$ or larger—the same order as the inflow ratio the rotor produces on its own. The yaw-rate contribution is therefore a first-order effect on the momentum balance, not a perturbation.

What survives, what changes. The momentum equation (25) is already written in terms of the composite V_∞; using (27) to include yaw rate requires no change of form. Likewise the BEMT thrust law (16) carries V_∞ symbolically and is unchanged in form by the yaw-rate contribution. What *is* changed is the operating point: the trim is no longer the hover trim $V_\infty = 0$ unless $\omega = 0$, so the hover-specialised forms (eqs. (26), (31), and the hover linearisation of Section 5) no longer apply during a sustained pirouette. The full axial or edgewise momentum closure must be used.

Effects not captured here. Three further consequences of yaw rate are second order at RC amplitudes and are not modelled in this note. (i) The hub-motion component also varies azimuthally with each blade's position—the classic "advancing/retreating" asymmetry of edgewise flight—but for pure yaw the hub motion is along the shaft, so this asymmetry is small. (ii) The disc itself has angular velocity ω about a vertical axis, which introduces small flapping and gyroscopic effects on the blades; these are outside the thrust-only model. (iii) The absolute rotor spin rate differs from the shaft-relative Ω_t by ω, a fractional change of order $\omega/\Omega_t \lesssim 1\%$ at the limits considered here, and is neglected.

3.4 Non-Dimensional Form

Dividing (25) through by $\rho A_t(\Omega_t R_t)^2$ converts it to a relation between non-dimensional groups. With the inflow ratios $\lambda_i = v_i/(\Omega_t R_t)$ and $\lambda = (V_\infty + v_i)/(\Omega_t R_t)$, and the edgewise advance ratio $\mu = V_\perp/(\Omega_t R_t)$, the result is

$$C_T \;=\; 2\,\lambda_i\,\sqrt{\mu^2+\lambda^2} \qquad \Longleftrightarrow \qquad \lambda_i \;=\; \frac{C_T}{2\,\sqrt{\mu^2+\lambda^2}}. \tag{28}$$

The square-root $\sqrt{\mu^2 + \lambda^2}$ is the non-dimensional resultant velocity at the disc: μ is the in-plane (edgewise) component, λ the through-disc component, and their hypotenuse is what the disc actually accelerates. Notice that $\lambda = \mu_z + \lambda_i$, where $\mu_z = V_\infty/(\Omega_t R_t)$. Therefore the right-hand expression in (28) is generally an *implicit* equation for λ_i, not a closed-form formula, because λ_i also appears inside λ.

Dimensional and non-dimensional forms in parallel. From here on we move between the dimensional induced velocity v_i and the inflow ratio λ_i freely; they encode the same physical quantity, related by $\lambda_i = v_i/(\Omega_t R_t)$. The dimensional form is convenient inside the dynamic-inflow ODE (Section 3.6), the non-dimensional form inside BEMT algebra, and the hover form v_h for sizing arguments. A useful reading rule is:

$$\lambda \quad \text{means total through-disc inflow, whereas} \quad \lambda_i \quad \text{means the part caused by the rotor itself.}$$

3.5 Closing the Coupled System

Equation (28) and the BEMT result (14) are two relations between the same three quantities (θ_t, C_T, λ_i, with $\lambda = \mu_z + \lambda_i$). For a given pitch θ_t they can be solved simultaneously for the steady-state inflow. In hover ($\mu = 0$ and $V_\infty = 0$, so $\lambda = \lambda_i$), the two equations reduce to

$$C_T \;=\; \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda_i}{2}\right), \qquad C_T \;=\; 2\lambda_i^2. \tag{29}$$

Equating them gives

$$2\lambda_i^2 \;=\; \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda_i}{2}\right) \qquad \Longrightarrow \qquad \lambda_i^2 + \frac{\sigma a}{8}\lambda_i - \frac{\sigma a}{12}\theta_t \;=\; 0. \tag{30}$$

This is a quadratic in λ_i. The two mathematical roots represent two possible signs of induced flow; for the positive-thrust hover branch we take the positive root:

$$\boxed{\lambda_i \;=\; \frac{\sigma a}{16}\left[\sqrt{1 + \frac{64\,\theta_t}{3\,\sigma a}} \;-\; 1\right] \quad (\text{hover, } \mu = 0).} \tag{31}$$

For small θ_t the bracket expands as $\frac{32\theta_t}{3\sigma a} - \mathcal{O}(\theta_t^2)$, giving $\lambda_i \approx 2\theta_t/3$—a useful order-of-magnitude check. It also explains a feature that can otherwise look strange: near zero pitch the BEMT term $\theta_t/3 - \lambda_i/2$ cancels to first order, so the hover thrust itself grows quadratically with pitch in this ideal lossless model. Real rotors add profile drag, blade stall limits, and finite root/tip corrections, so this exact small-pitch behaviour should not be over-interpreted.

3.6 Wake Inertia

Equation (31) gives the *steady* inflow for a given pitch, but it ignores time. In reality the wake is a column of air with mass, and changing the rotor's thrust requires accelerating that column—which cannot happen instantaneously. If at time $t = 0$ the pilot steps the pitch up, the blade-element thrust jumps immediately to its frozen-wake value (the wake hasn't moved yet, so the local angle of attack is still $\theta_t - \phi$ at the old ϕ). Over the next few milliseconds the induced velocity builds toward its new steady value, the local angle of attack drops, and the thrust relaxes to its lower equilibrium. The rotor exhibits a *lead–lag* response in thrust because the inflow has *lag*.

For control-design purposes the simplest faithful model treats the uniform component of the induced velocity as a first-order state:

$$\boxed{\tau_i\,\dot{v}_i + v_i \;=\; v_{i,\mathrm{ss}},} \tag{32}$$

where $v_{i,\mathrm{ss}}$ is the steady-state induced velocity that solves (25) for the *current* thrust T_t, and τ_i is the wake-relaxation time constant. This is the *one-state Pitt–Peters* dynamic inflow model. Equation (32) is the inflow analogue of an RC low-pass filter: $v_{i,\mathrm{ss}}$ is the target, v_i is the state, and τ_i sets how quickly the state chases the target.

The notation $v_{i,\mathrm{ss}}$ deserves care. It does not mean a new physical velocity in the wake; it means "the value that v_i would eventually settle to if the present thrust and flight condition were held fixed." For example, in hover $v_{i,\mathrm{ss}} = \sqrt{T_t/(2\rho A_t)}$. In edgewise or axial flow, $v_{i,\mathrm{ss}}$ is defined implicitly by

$$T_t \;=\; 2\rho A_t\, v_{i,\mathrm{ss}} \sqrt{(V_\infty + v_{i,\mathrm{ss}})^2 + V_\perp^2}. \tag{33}$$

The dynamic model says that the real induced velocity $v_i(t)$ lags behind this steady target.

3.7 The Pitt–Peters Time Constant

What sets τ_i? Two mechanisms contribute. The first is the simple flow-traversal time: an air parcel takes roughly R_t/v_h to cross the rotor disc, so τ_i must scale this way on dimensional grounds alone. The second, more subtle effect is the *apparent mass* of the air column: when an impermeable disc is accelerated through fluid, the fluid in its immediate neighbourhood is accelerated with it, contributing an extra inertia even in the absence of a true rotor wake. Lamb's classical potential-flow result gives the apparent mass of a thin circular disc as $\rho\,(8/3)\,R_t^3$, a fraction $8/(3\pi) \approx 0.849$ of the mass of a sphere of radius R_t. Pitt and Peters identified this same coefficient as governing the lag of the uniform-inflow component, yielding

$$\boxed{\tau_i \;=\; \frac{8/(3\pi)}{4\,\Omega_t\,\lambda_i^{\mathrm{trim}}} \;=\; \frac{0.849\,R_t}{4\,v_h},} \tag{34}$$

where the two forms are equivalent because, at trim,

$$\lambda_i^{\mathrm{trim}} = \frac{v_h}{\Omega_t R_t}, \qquad \Omega_t\,\lambda_i^{\mathrm{trim}} R_t = v_h. \tag{35}$$

The right-hand form is the easier to interpret: τ_i is essentially the time for an air parcel travelling at the hover induced velocity to cross a quarter of the disc radius, scaled by the apparent-mass coefficient. Higher disc loading $\Rightarrow$ faster v_h $\Rightarrow$ shorter τ_i: hard-working rotors have snappier wakes.

In (34), $\lambda_i^{\mathrm{trim}}$ and v_h are trim values, not instantaneous signals. This keeps the one-state model a linear first-order lag around the selected operating point. For a wide-envelope implementation one may update τ_i slowly from the current or commanded thrust, but it should be bounded away from zero thrust because the formula becomes singular as $v_h \to 0$.

Caveat on Pitt–Peters time constants. The closed-form (34) is known to underpredict measured inflow lag on full-scale rotors by a factor of 2–4. For RC tail rotors it likely underpredicts somewhat less, but the value should be regarded as a lower bound; see Section 8.

3.8 The Coupled Model

Combining the blade-element thrust relation with the dynamic-inflow state gives the complete tail-rotor model of this note:

$$\boxed{\begin{aligned}
&T_t = \rho A_t(\Omega_t R_t)^2\,\frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{v_i + V_\infty}{2\Omega_t R_t}\right), && (36)\\
&T_t = 2\rho A_t\, v_{i,\mathrm{ss}}\sqrt{(V_\infty + v_{i,\mathrm{ss}})^2 + V_\perp^2},\\
&\dot v_i = \frac{1}{\tau_i}\big(v_{i,\mathrm{ss}} - v_i\big). && (37)
\end{aligned}}$$

Equation (36) is BEMT, algebraic in θ_t and v_i. Equation (37) is Pitt–Peters: a first-order ODE in v_i driven by the instantaneous thrust through the momentum-theory steady-state map. In hover this ODE can be written explicitly as $\dot{v}_i = (\sqrt{T_t/(2\rho A_t)} - v_i)/\tau_i$; away from hover the definition of $v_{i,\mathrm{ss}}$ is implicit and is normally solved by a small scalar iteration or by using the equivalent residual form in software. The pair is a one-state nonlinear system with input θ_t and output T_t, parameterised by the flight condition $(V_\infty, V_\perp)$ and the rotor constants. This pair is inverted in Section 4, analysed in the frequency domain—and characterised across the operating envelope—in Section 5, and packaged for implementation in Section 7.

4 Inverse Problem: Blade Pitch from Commanded Thrust

The coupled model of Section 3—blade-element thrust (36) and dynamic inflow (37)—maps a pitch input θ_t to a thrust output T_t. For control design we need to run this map *backwards*: given a commanded thrust $T_t^{\mathrm{cmd}}(t)$, what pitch $\theta_t(t)$ should we apply? The answer will sit inside the tail-rotor inner loop as a feedforward block: the outer loop (yaw control) produces a thrust demand, the inverse block translates it into pitch, and the servo drives the tail-rotor pitch mechanism.

This section derives the inverse in two steps. First, we invert the blade-element relation algebraically to obtain the *instantaneous* pitch–thrust inverse, assuming the induced velocity is known (Section 4.1). Second, we expand to the dynamic case by carrying the induced velocity as a state, governed by the Pitt–Peters ODE (Section 4.2). The result is a complete nonlinear dynamic inversion of the rotor plant. Section 5 linearises the system about hover trim, characterises its lead–lag response in the frequency domain, and quantifies the resulting inflow-softening factor across the operating envelope; Section 7 then collects the inverse into a compact form for direct implementation in an inner control loop.

It is useful to separate two ideas before doing the algebra. The blade element relation is an *instantaneous map*: at the present value of v_i, it tells us what thrust a chosen pitch will produce right now. The inflow equation is a *state update*: it tells us how v_i will move after the thrust command is applied. The inverse therefore does not predict a future wake and then choose pitch from that future wake. It uses the current wake estimate to choose the present pitch, while the Pitt–Peters equation advances the wake estimate for the next instant.

4.1 The Instantaneous Inverse (Algebraic Step)

The blade-element relation (16) is linear in $(\theta_t, v_i, V_\infty)$ at fixed Ω_t, and therefore trivially invertible for θ_t. Solving (16) for the pitch and setting the desired thrust to T_t^{cmd} gives

$$\boxed{\theta_t \;=\; \frac{6\,T_t^{\mathrm{cmd}}}{\rho A_t(\Omega_t R_t)^2\,\sigma a} \;+\; \frac{3\,(v_i + V_\infty)}{2\,\Omega_t R_t}.} \tag{38}$$

This is the *exact* inverse at every instant, conditional on knowing the current v_i and V_∞: apply pitch (38) and the blade element will, instantaneously, produce thrust T_t^{cmd}.

Two terms appear on the right-hand side. The first is the pitch the rotor would need if no induced inflow were present—a naive "pitch per unit thrust" calculation, with denominator $\rho A_t(\Omega_t R_t)^2\,\sigma a/6$ (the frozen-wake thrust-per-radian gain from Section 2.6). The second is the extra pitch needed to overcome inflow's stolen incidence: v_i and V_∞ together set the inflow angle ϕ, and the blade must be pitched up by the kinematic factor $3/(2\Omega_t R_t)$ of (17) just to retain its original effective angle of attack α. Equation (38) is, in essence, the control-theoretic restatement of the kinematic story from Section 2.

The word *instantaneous* is important. The pitch in (38) compensates the inflow that exists now; it does not use the steady-state inflow that will exist after the wake has settled. Immediately after a thrust step, v_i is still close to its old value, so the pitch command initially looks like a frozen-wake inverse. As v_i evolves, the same formula automatically adds the extra pitch needed to hold the commanded thrust against the stronger induced flow.

4.2 Expanding to the Dynamic System

Equation (38) requires the *current* value of v_i. On an RC platform v_i is not directly measurable: there is no sensor that reports the induced velocity through the disc. The inverse must therefore carry an internal estimate of v_i, propagated by the same physics that governs the real rotor's inflow— namely, the Pitt–Peters ODE (32) closed by the momentum relation (25). The commanded thrust first defines the steady inflow target $v_{i,\mathrm{ss}}$:

$$T_t^{\mathrm{cmd}} = 2\rho A_t\, v_{i,\mathrm{ss}} \sqrt{(V_\infty + v_{i,\mathrm{ss}})^2 + V_\perp^2}, \tag{39}$$

$$\dot{v}_i = \frac{1}{\tau_i}\big(v_{i,\mathrm{ss}} - v_i\big), \tag{40}$$

$$\tau_i = \frac{0.849}{4\,\Omega_t \lambda_i}, \qquad \lambda_i = v_i/(\Omega_t R_t). \tag{41}$$

Equation (39) is the momentum closure: it asks, "if the rotor were held at the commanded thrust and the flight condition were fixed, what induced velocity would the wake settle to?" In hover this has the explicit solution $v_{i,\mathrm{ss}} = \sqrt{T_t^{\mathrm{cmd}}/(2\rho A_t)}$. With axial or edgewise flow it is a scalar nonlinear equation for $v_{i,\mathrm{ss}}$; the positive working-state root is the one used here. Equation (40) is then the wake's relaxation: the current induced velocity chases that target on the timescale τ_i. Equation (41) is the apparent-mass time constant from Section 3, written in a scheduled form using the current inflow estimate. A fixed-trim implementation can instead hold τ_i constant at the scheduled trim value.

Substituting the algebraic inverse (38) for θ_t and using the dynamic state v_i from (39)–(41), we have a complete *nonlinear dynamic inversion*: a one-state nonlinear system whose input is the commanded thrust and whose output is the pitch command. No linearisation is involved; no operating point has been fixed; no small-signal assumption has been made. The model is exact within the BEMT and Pitt–Peters assumptions inherited from Sections 2–3.

A subtle point: $v_{i,\mathrm{ss}}$ in (39) appears to be driven by T_t^{cmd} (what we want), not by T_t (what we produce). At a control-theoretic level these are the same if the inverse is working—θ_t is computed precisely so that $T_t = T_t^{\mathrm{cmd}}$ at the blade element—so driving the inflow estimator from T_t^{cmd} is self-consistent and avoids needing a thrust measurement. The choice does become relevant when servo or sensor delays are introduced, but that is outside this section's scope.

Putting the pieces together, the continuous-time inverse has a simple causal structure:

$$\boxed{\text{current } v_i,\ T_t^{\mathrm{cmd}},\ V_\infty,\ V_\perp} \quad\longrightarrow\quad \boxed{\theta_t \text{ from (38)}} \quad \text{and} \quad \boxed{\dot{v}_i \text{ from (39)–(41)}}.$$

The pitch output uses the current state; the state derivative prepares the wake estimate used by the next control instant. This is the same division of labour used in the implementation form of Section 7.

5 Transfer Function Analysis

The nonlinear plant (36)–(37) of Section 3—and its inverse derived in Section 4.2—can be analysed in the frequency domain by linearising about a chosen trim point. The resulting transfer function exposes the pole and zero structure of the rotor, the "lead–lag" character of its thrust response to a step in pitch, and the inflow-softening factor that quantifies how much DC gain is lost to the wake. This section closes by showing how this softening factor varies across the operating envelope. The linearised model is *not* an alternative controller—the inner loop will run the full nonlinear inverse, packaged for implementation in Section 7—but it gives the language in which the plant's bandwidth and stability margins are usually discussed, and is essential when an outer loop must be designed around this plant.

5.1 Linearisation About Hover Trim

Choosing trim. By *trim* we mean an equilibrium operating point of the plant: a constant pitch $\bar{\theta}$ producing a constant thrust $\bar{T}$ with a constant induced velocity $\bar{v}_i$, all satisfying both BEMT and momentum theory simultaneously. For a tail rotor on a hovering helicopter the natural choice is the trim that balances main-rotor anti-torque in still air. We specialise to hover ($V_\infty = V_\perp = 0$) because the algebra is cleanest and because the tail rotor spends most of its operating life near this condition; the non-hover case generalises by replacing $\bar{v}_i$ with the hover-equivalent $v_h = \sqrt{\bar{T}/(2\rho A_t)}$ and adjusting the trim accordingly.

Perturbation convention. Write each variable as a constant trim value plus a small perturbation:

$$\theta_t = \bar{\theta} + \delta\theta_t, \quad v_i = \bar{v}_i + \delta v_i, \quad T_t = \bar{T} + \delta T. \tag{42}$$

We retain terms linear in the δ-quantities and discard products of two perturbations (assumed small $\times$ small). The trim values themselves satisfy the steady model and cancel out of the perturbation equations, leaving a system in the δ-variables alone.

What needs linearising. The blade-element relation (16) is *already linear* in its arguments at fixed Ω_t; reading off its coefficients gives the perturbation form

$$\delta T \;=\; G_0\,\delta\theta_t \;-\; \frac{3\,G_0}{2\Omega_t R_t}\,\delta v_i, \qquad G_0 \;\equiv\; \frac{\rho A_t (\Omega_t R_t)^2\,\sigma a}{6}, \tag{43}$$

where G_0 is the frozen-wake thrust-per-radian gain: the high-frequency (instantaneous) response of thrust to a step in pitch before the wake has time to react. The pitch and inflow coefficients inherit the kinematic factor (17) that separates them. The nonlinearity is entirely in the momentum relation $\bar{T} = 2\rho A_t \bar{v}_i^2$, which couples thrust to induced velocity through a square. Differentiating implicitly,

$$\left.\frac{\partial T}{\partial v_i}\right|_{\text{trim}} \;=\; 4\,\rho A_t\,\bar{v}_i, \qquad \Longrightarrow \qquad \delta v_{i,\text{ss}} \;=\; \frac{1}{4\rho A_t \bar{v}_i}\,\delta T \;\equiv\; k_v\,\delta T, \tag{44}$$

where k_v is the linearised sensitivity of steady-state induced velocity to thrust at the chosen trim:

$$k_v \;\equiv\; \left.\frac{\partial v_{i,\text{ss}}}{\partial T}\right|_{\text{trim}} \;=\; \frac{1}{4\rho A_t \bar{v}_i} \;=\; \frac{\bar{v}_i}{2\bar{T}}, \qquad \text{units: m}\,\text{s}^{-1}\,\text{N}^{-1}. \tag{45}$$

The two forms come from the same expression via the hover relation $\bar{T} = 2\rho A_t \bar{v}_i^2$. Physically: k_v tells us *how much extra induced velocity each newton of additional thrust will pull through the disc, in steady state.* It is large for low-thrust trims (where $\bar{v}_i$ is small and any new thrust must accelerate a slow wake from near-rest) and small for high-thrust trims (where $\bar{v}_i$ is already substantial and incremental thrust matters less proportionally).

Linearised inflow dynamics. The Pitt–Peters ODE (32) is already linear in v_i at fixed τ_i. Substituting $v_{i,\mathrm{ss}} = \bar{v}_i + k_v\,\delta T$ and subtracting the trim equation $\bar{v}_i = \bar{v}_i$ leaves

$$\tau_i\,\delta\dot{v}_i + \delta v_i \;=\; k_v\,\delta T. \tag{46}$$

Taking the Laplace transform with zero initial conditions ($(\dot{\cdot}) \to s\,(\cdot)$):

$$(\tau_i s + 1)\,\delta v_i(s) \;=\; k_v\,\delta T(s). \tag{47}$$

5.2 The Linearised Plant Transfer Function

Equations (43) and (47) are a two-equation linear system in the three Laplace-domain quantities $\delta\theta_t(s)$, $\delta T(s)$, $\delta v_i(s)$. Eliminate δv_i: from (47),

$$\delta v_i(s) \;=\; \frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{48}$$

Substitute into (43):

$$\delta T(s) \;=\; G_0\,\delta\theta_t(s) \;-\; \frac{3\,G_0}{2\Omega_t R_t}\,\frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{49}$$

Collect δT, defining the dimensionless inflow–thrust coupling that emerges from the algebra:

$$\beta \;\equiv\; \frac{3\,G_0\,k_v}{2\,\Omega_t R_t} \;=\; \frac{\sigma a}{16\,\lambda_h}, \tag{50}$$

where the closed-form right-hand expression follows on substituting $G_0 = \rho A_t(\Omega_t R_t)^2\sigma a/6$ and the hover-trim sensitivity $k_v = 1/(4\rho A_t \bar{v}_i)$ from (44), together with $\lambda_h = \bar{v}_i/(\Omega_t R_t)$. Then

$$\delta T(s)\left[1 + \frac{\beta}{\tau_i s + 1}\right] \;=\; G_0\,\delta\theta_t(s), \tag{51}$$

and rearranging to standard form,

$$\boxed{\frac{\delta T(s)}{\delta\theta_t(s)} \;=\; \frac{G_0\,(\tau_i s + 1)}{\tau_i s + (1+\beta)}.} \tag{52}$$

This is the linearised tail-rotor plant: thrust per radian of pitch perturbation about hover trim, valid for small $\delta\theta_t$ over a bandwidth limited by the linearisation.

5.3 Reading the Plant: Lead–Lag Structure

The plant (52) has one zero and one pole, both on the negative real axis:

$$\text{zero at } s = -\frac{1}{\tau_i}, \qquad \text{pole at } s = -\frac{1+\beta}{\tau_i}. \tag{53}$$

Because $\beta > 0$, the pole sits at a higher frequency than the zero. The high-frequency and DC limits of the gain are

$$\lim_{s\to\infty} \frac{\delta T}{\delta\theta_t} = G_0 \qquad \text{(instantaneous, “open-loop” gain),} \tag{54}$$

$$\lim_{s\to 0} \frac{\delta T}{\delta\theta_t} = \frac{G_0}{1+\beta} \qquad \text{(steady-state, “closed-wake” gain).} \tag{55}$$

The DC gain is lower than the instantaneous gain by a factor $1+\beta$. A step in pitch produces an immediate thrust spike of size $G_0\,\delta\theta_t$ (before the wake responds), which then relaxes over a timescale $\tau_i/(1+\beta)$ to the lower steady value $G_0\,\delta\theta_t/(1+\beta)$. This is the *lead–lag* response described qualitatively in the introduction, now in closed form.

Physical reading of β. The coupling β of (50) measures how strongly induced inflow couples back into the thrust. Tracing its factors: the kinematic ratio $3/(2\Omega_t R_t)$ is how much thrust each m/s of induced velocity steals per radian of frozen-wake gain; k_v is how much extra induced velocity each newton of thrust pulls through the disc. Their product closes the loop and gives the fractional thrust loss between the instantaneous and steady-state response:

$$\frac{\text{DC gain}}{\text{instantaneous gain}} = \frac{1}{1+\beta} = \frac{1}{1+\sigma a/(16\lambda_h)}. \tag{56}$$

For typical RC tail rotors at hover trim, β sits in the range 0.5–1.5—meaning the DC gain is anywhere from 40% to 70% of the open-loop gain. The thrust the rotor delivers in steady state is significantly less than what its blade pitch suggests. This phenomenon is *inflow softening*, and Section 5.5 rederives the same coupling $\sigma a/(16\lambda)$ directly from the closed BEMT–momentum system, confirming (50) and extending it across the operating envelope.

5.4 Inverting the Plant: The Lead Compensator

A feedforward compensator that cancels the plant dynamics is, by definition, the inverse of the plant. Reciprocating (52):

$$\boxed{\frac{\delta\theta_t(s)}{\delta T(s)} = \frac{\tau_i s + (1+\beta)}{G_0\,(\tau_i s + 1)}}. \tag{57}$$

The roles of pole and zero are now swapped: the compensator has its *zero* where the plant had its *pole*, and vice versa. The compensator is, in classical control terminology, a *lead network*: its zero sits at a lower frequency than its pole, so its phase contribution is positive across the intervening band. Cascading (57) $\rightarrow$ (52) gives the identity transfer function. This is the linearised counterpart of the nonlinear dynamic inverse derived in Section 4.2: cancellation through the wake dynamics, expressed in the frequency domain rather than as a state-update.

5.5 The Inflow-Softening Factor

The DC gain $G_0/(1+\beta)$ varies with the trim point because $k_v \propto 1/\bar{v}_i \propto 1/\sqrt{\bar{T}}$. The factor $(1+\beta)$ is the *inflow softening denominator*; softening is strongest at low thrust.

The coupling β was identified in (50) by combining the BEMT perturbation with the momentum-derived sensitivity k_v. The same coupling can be derived more directly, without ever linearising about an explicit trim, by closing the coupled BEMT–momentum system in hover and differentiating the resulting implicit relation. The two routes agree, as they must, and the second makes the trim consistency manifest. In hover, the coupled equations are

$$C_T = \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \tag{BEMT}$$

$$\lambda = \sqrt{C_T/2}. \tag{Momentum}$$

Eliminating C_T:

$$\lambda^2 = \frac{\sigma a}{4}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \tag{58}$$

with closed form

$$\lambda(\theta_t) = \frac{\sigma a}{16}\left[\sqrt{1 + \frac{64\theta_t}{3\sigma a}} - 1\right]. \tag{59}$$

Trim sensitivity. Differentiating implicitly:

$$2\lambda\,\frac{d\lambda}{d\theta_t} = \frac{\sigma a}{4}\left(\frac{1}{3} - \frac{1}{2}\frac{d\lambda}{d\theta_t}\right) \quad\Longrightarrow\quad \frac{d\lambda}{d\theta_t} = \frac{\sigma a}{24\lambda + 3\sigma a/2}. \tag{60}$$

Then

$$\boxed{\left.\frac{dC_T}{d\theta_t}\right|_{\text{steady}} = \frac{\sigma a}{6}\cdot\underbrace{\frac{1}{1+\dfrac{\sigma a}{16\lambda}}}_{S(\lambda)}.} \tag{61}$$

The factor $\sigma a/6$ is the instantaneous (high-frequency) gain (non-dimensional form of G_0); $S(\lambda)$ is the softening factor. Comparing with the linearised form $S = 1/(1+\beta)$ from Section 5.3 confirms the closed form already given in (50):

$$\boxed{\beta \;=\; \frac{\sigma a}{16\,\lambda_h}, \qquad \lambda_h = \sqrt{C_T/2}.} \tag{62}$$

This is the expression to use in design. It depends only on the aerodynamic group σa and the trim inflow ratio λ_h—no separate evaluations of dimensional gains are needed.

5.6 Magnitude Across the Operating Envelope

For a 700 class tail rotor ($\sigma = 0.136$, $a = 5.7$, hover $\bar{C}_T = 0.0076$, $\bar{\lambda} = 0.062$):

Thrust	C_T	λ	$\beta = \sigma a/(16\lambda)$	$S(\lambda)$
25% hover	0.0019	0.031	1.58	0.39
50% hover	0.0038	0.044	1.12	0.47
Hover	0.0076	0.062	0.79	0.56
150% hover	0.0113	0.075	0.65	0.61
200% hover	0.0151	0.087	0.56	0.64

The DC gain varies by roughly 40% across the operating envelope. A fixed-gain lead compensator will mis-track by this amount at the extremes.

Time constant varies too. The Pitt–Peters time constant $\tau_i = 0.849/(4\Omega_t\lambda)$ scales as $1/\lambda \propto 1/\sqrt{C_T}$. For the same 700 class rotor:

Thrust	τ_i [ms]
25% hover	6.7
Hover	3.3
200% hover	2.4

Both the gain and the time constant of the rotor depend on the trim operating point, and both must be scheduled for a fixed-coefficient controller to track across the envelope.

6 Non-Dimensional Inverse and Minimum Parameter Set

The dynamic inverse derived in Section 4 is written in dimensional form: its inputs are a commanded thrust in newtons, axial and edgewise flows in m/s, and a yaw rate in rad/s; its output is a pitch angle in radians; its internal state v_i is a velocity. In an actual flight-control stack, however, the upstream yaw PID does not deliver a thrust in newtons. It delivers a unitless command—typically in a normalised range like $[0, 1]$ or $[-1, 1]$—representing a fraction of the available yaw authority. The thrust appears only as a fiction inside the inverse, used to drive the BEMT and momentum algebra.

This section recasts the inverse in dimensionless form. The recipe is standard. First, pick one reference scale—here, a reference thrust T_{ref}; every other quantity inherits a reference value from it by the same physics that defined it in the dimensional inverse (Section 6.1). Substituting these scaled forms into each equation of the inverse, the dimensional constants cancel and the inverse collapses to three update equations carrying just four dimensionless coefficients $(K_u, K_v, K_\tau, K_\omega)$ (Section 6.2). Those four coefficients in turn depend on only a handful of size-invariant dimensionless groups; those groups, not the coefficients themselves, are what the firmware exposes to the user—see Section 7.1 in the implementation chapter, where the mapping from exposed parameters to $(K_u, K_v, K_\tau, K_\omega)$ is spelled out. The dimensionless update law below is the one an embedded controller stores and runs each tick; Section 7 wraps it in the I/O, state, and simplifications a practitioner needs.

6.1 Reference Scales

Choose a reference thrust T_{ref}: the thrust to which the dimensionless command $u = 1$ corresponds. The choice is the implementer's; sensible candidates include the hover trim thrust $\bar{T}$, or some multiple of it that brackets the available 3D authority. Once T_{ref} is fixed, every other reference follows from it. The hover induced velocity that would correspond to this reference thrust is

$$v_{\text{ref}} \equiv \sqrt{\frac{T_{\text{ref}}}{2\rho A_t}}, \tag{63}$$

and the corresponding inflow ratio is

$$\lambda_{\text{ref}} \equiv \frac{v_{\text{ref}}}{\Omega_t R_t}. \tag{64}$$

With these two scales fixed, every dimensional quantity in the inverse has a natural dimensionless partner:

$$u \equiv \frac{T_t^{\text{cmd}}}{T_{\text{ref}}}, \quad \hat{v}_i \equiv \frac{v_i}{v_{\text{ref}}}, \quad \hat{v}_{i,\text{ss}} \equiv \frac{v_{i,\text{ss}}}{v_{\text{ref}}}, \quad \hat{V}_\infty \equiv \frac{V_\infty}{v_{\text{ref}}}, \quad \hat{V}_\perp \equiv \frac{V_\perp}{v_{\text{ref}}}. \tag{65}$$

The pitch θ_t is already dimensionless (radians) and is carried unchanged. Time can also be made dimensionless by referring it to the characteristic wake timescale

$$\tau_{\text{ref}} \equiv \frac{0.849\, R_t}{4\, v_{\text{ref}}}, \tag{66}$$

which is the Pitt–Peters time constant from (34) evaluated at $v_i = v_{\text{ref}}$.

6.2 The Dimensionless Update Law

Substituting (65) into the three equations of the dynamic inverse—(38), (39), and (40)–(41)—each one simplifies.

Momentum closure. Equation (39) reads $T_t^{\mathrm{cmd}} = 2\rho A_t\, v_{i,\mathrm{ss}}\sqrt{(V_\infty + v_{i,\mathrm{ss}})^2 + V_\perp^2}$. Substituting $T_t^{\mathrm{cmd}} = T_{\mathrm{ref}} u = 2\rho A_t v_{\mathrm{ref}}^2\, u$ (by (63)) and the velocity scalings, the prefactor $2\rho A_t v_{\mathrm{ref}}^2 = T_{\mathrm{ref}}$ cancels from both sides:

$$\boxed{u \;=\; \hat{v}_{i,\mathrm{ss}}\sqrt{(\hat{V}_\infty + \hat{v}_{i,\mathrm{ss}})^2 + \hat{V}_\perp^2}.} \tag{67}$$

No rotor parameter appears: this is the same momentum closure for every rotor. In hover ($\hat{V}_\infty = \hat{V}_\perp = 0$) the explicit solution is $\hat{v}_{i,\mathrm{ss}} = \sqrt{u}$, the dimensionless analogue of the hover formula (26).

Pitch from command and state. The instantaneous inverse (38) reads

$$\theta_t \;=\; \frac{6}{\rho A_t (\Omega_t R_t)^2\, \sigma a}\, T_t^{\mathrm{cmd}} \;+\; \frac{3}{2\,\Omega_t R_t}\,(v_i + V_\infty).$$

Substituting $T_t^{\mathrm{cmd}} = T_{\mathrm{ref}} u = 2\rho A_t v_{\mathrm{ref}}^2\, u$ and the velocity scalings, and using $\lambda_{\mathrm{ref}} = v_{\mathrm{ref}}/(\Omega_t R_t)$:

$$\theta_t \;=\; \frac{12\,\lambda_{\mathrm{ref}}^2}{\sigma a}\, u \;+\; \frac{3\,\lambda_{\mathrm{ref}}}{2}\,(\hat{v}_i + \hat{V}_\infty). \tag{68}$$

Two dimensionless gains emerge:

$$\boxed{\begin{aligned} K_u \;&\equiv\; \frac{12\,\lambda_{\mathrm{ref}}^2}{\sigma a}, && (69)\\ K_v \;&\equiv\; \frac{3\,\lambda_{\mathrm{ref}}}{2}, && (70)\end{aligned}}$$

giving the dimensionless pitch law

$$\theta_t \;=\; K_u\, u \;+\; K_v\,(\hat{v}_i + \hat{V}_\infty). \tag{71}$$

K_u is the pitch (in radians) demanded per unit command when the state and free-stream contributions are zero; K_v is the pitch demanded per unit dimensionless through-disc velocity. The right-hand expressions in (69)–(70) contain only λ_{ref} and σa; air density, disc area, tip speed, and every other dimensional constant of the rotor have cancelled. Moreover, since both gains are built from the same two ingredients, they are not independent: eliminating λ_{ref} between (69) and (70) yields

$$\frac{K_u}{K_v^2} \;=\; \frac{16}{3\,\sigma a}, \tag{72}$$

so the pair (K_u, K_v) carries exactly the information of $(\lambda_{\mathrm{ref}}, \sigma a)$—one inflow-ratio number and one blade-loading number.

Inflow state update. The Pitt–Peters ODE (40) with the state-scheduled time constant (41) reads $\dot{v}_i = (v_{i,\mathrm{ss}} - v_i)/\tau_i$ with $\tau_i = 0.849 R_t/(4 v_i)$. Substituting $v_i = v_{\mathrm{ref}}\hat{v}_i$, $v_{i,\mathrm{ss}} = v_{\mathrm{ref}}\hat{v}_{i,\mathrm{ss}}$, and $\tau_i = \tau_{\mathrm{ref}}/\hat{v}_i$ by (66), the factor v_{ref} cancels from both sides of the time derivative and we are left with

$$\tau_{\mathrm{ref}}\,\dot{\hat{v}}_i \;=\; \hat{v}_i\,(\hat{v}_{i,\mathrm{ss}} - \hat{v}_i). \tag{73}$$

If time is measured in units of τ_{ref} (i.e. $\tilde{t} \equiv t/\tau_{\mathrm{ref}}$) this equation becomes fully parameter-free. Discretising with a control-loop period Δt introduces a single dimensionless gain

$$\boxed{K_\tau \;\equiv\; \frac{\Delta t}{\tau_{\mathrm{ref}}} \;=\; \frac{4\, v_{\mathrm{ref}}\,\Delta t}{0.849\, R_t},} \tag{74}$$

giving the explicit-Euler update

$$\hat{v}_i \;\leftarrow\; \hat{v}_i \;+\; K_\tau\, \hat{v}_i\,(\hat{v}_{i,\mathrm{ss}} - \hat{v}_i). \tag{75}$$

K_τ is the only place the control-loop period Δt and the rotor radius R_t enter the inverse.

Yaw-rate contribution. Finally, the axial-flow composition $V_\infty = V_\infty^{\text{aero}} + \ell_t \omega$ from (3), divided by v_{ref}, absorbs the geometric ℓ_t into a scale factor with units of seconds

$$\boxed{K_\omega \equiv \frac{\ell_t}{v_{\text{ref}}},} \tag{76}$$

giving the dimensionless yaw-rate scaling

$$\hat{V}_\infty = \hat{V}_\infty^{\text{aero}} + K_\omega \,\omega. \tag{77}$$

The airframe airspeed component $\hat{V}_\infty^{\text{aero}}$ either arrives pre-non-dimensionalised from the airspeed estimator or is set to zero for near-hover use.

Summary. The complete dimensionless inverse is the three update equations (67), (71), and (75), with the yaw-rate scaling (77) on the input side. The stored constants are

$$K_u, \quad K_v, \quad K_\tau, \quad K_\omega \tag{78}$$

plus any constants used to non-dimensionalise the airspeed estimator's output (typically zero on tail rotors without airspeed). Every other physical parameter of the rotor and air has dropped out. Counting independent content: K_u and K_v together carry two numbers (λ_{ref} and σa, via (72)); K_τ brings the rotor radius and the loop period; K_ω brings the moment arm.

The steady command–pitch curve. A useful by-product of the dimensionless pitch law is the steady response in pure hover. Setting $\hat{v}_i = \hat{v}_{i,\text{ss}} = \sqrt{u}$ and $\hat{V}_\infty = 0$ in (71) gives

$$\theta_t^{\text{steady}}(u) = K_u \, u + K_v \sqrt{u}, \tag{79}$$

which depends on (K_u, K_v) alone—K_τ only enters in transients, K_ω only when the airframe yaws.

7 Control Loop Implementation

The instantaneous inverse (38) together with the inflow model (39)–(41) is a complete nonlinear dynamic inversion of the tail-rotor plant. This section collects those equations into a compact form suitable for direct implementation in an inner control loop: first the small set of exposed parameters a builder configures and the initialisation-time map from those parameters to the four stored coefficients $(K_u, K_v, K_\tau, K_\omega)$ (Section 7.1); then the live-tick separation of fixed coefficients, inputs, state, and outputs (Section 7.2); then the update law in the form a practitioner would actually code (Section 7.3); and finally the simplifications a constrained embedded environment will typically apply (Section 7.4). The transfer-function analysis of Section 5—including the envelope behaviour of the softening factor—informs how the resulting controller will behave across the thrust range, but is not a prerequisite for running the loop itself.

7.1 Exposed Parameters

Section 6 reduced the loop to four dimensionless coefficients $(K_u, K_v, K_\tau, K_\omega)$, but the firmware still has to choose those numbers. Rather than expose the coefficients directly, we expose a smaller set of more physically intuitive quantities and compute $(K_u, K_v, K_\tau, K_\omega)$ from them at initialisation time.

We fix the implementer's choice of reference scale to $T_{\text{ref}} = \bar{T}$, so that the dimensionless command $u = 1$ corresponds to hover trim. Then $\lambda_{\text{ref}} = \lambda_h$ and $v_{\text{ref}} = \bar{v}_i$, and the exposed parameters are three dimensionless groups:

- $\bar{\theta}$ — hover trim pitch. The tail-rotor blade pitch measured at hover with the heli's nominal loadout, in degrees, at the nominal governed headspeed Ω_t. Directly visible on telemetry. The knob that distinguishes a gently-loaded rotor from an aggressively-loaded one, replacing any intermediate aerodynamic quantity that would otherwise have to be computed or assumed.

- σa — blade loading parameter. Lift produced per radian of pitch, normalised by disc area. Default $\sigma a \approx 0.8$ for a two-bladed symmetric tail rotor; adjustment is only needed for non-standard blades.

- ℓ_t/R_t — boom-to-tail-radius ratio. How far behind the airframe yaw centre the tail sits, in tail radii. Default $\ell_t/R_t \approx 7$.

Typical ranges across the three RC size classes of Section 8, at hover trim:

Quantity	420 mm	550 mm	700 mm
$\bar{\theta}$ [deg]	4–10	5–12	5–12
σa	0.6–1.0	0.7–1.0	0.7–0.85
ℓ_t/R_t	4.5–7.5	5.4–8.0	5.9–7.4

The bands overlap heavily across all three classes; each exposed parameter varies by under a factor of three across the envelope. $\bar{\theta}$ is directly observable on telemetry, σa is nearly fixed by blade type (two-bladed symmetric foil), and ℓ_t/R_t by airframe convention—so in practice only $\bar{\theta}$ is set per flight, the other two from their defaults.

Mapping to the stored coefficients. With $T_{\text{ref}} = \bar{T}$, the reference scales of Section 6.1 collapse onto hover trim: $\lambda_{\text{ref}} = \lambda_h$ and $v_{\text{ref}} = \bar{v}_i = \lambda_h \, \Omega_t \, R_t$. The four coefficients (69), (70), (74), (76) of Section 6.2 then depend on a single rotor scalar λ_h, given σa, ℓ_t/R_t, Ω_t, and the loop period Δt. The firmware does not measure λ_h directly; it derives it from $\bar{\theta}$ by inverting the steady command–pitch curve (79) at $u = 1$, which is exactly $K_u + K_v = \bar{\theta}$. Substituting

(69)–(70) yields the quadratic $(12/\sigma a)\,\lambda_h^2 + (3/2)\,\lambda_h - \bar{\theta} = 0$, whose positive root is

$$\lambda_h \;=\; \frac{\sigma a}{24}\left(-\frac{3}{2} \;+\; \sqrt{\frac{9}{4} + \frac{48\,\bar{\theta}}{\sigma a}} \right). \tag{80}$$

The stored coefficients then follow by substituting $\lambda_{\mathrm{ref}} = \lambda_h$ and $v_{\mathrm{ref}} = \lambda_h\,\Omega_t\,R_t$ into (69)–(76):

$$K_u = \frac{12\,\lambda_h^2}{\sigma a}, \quad K_v = \frac{3\,\lambda_h}{2}, \quad K_\tau = \frac{\Delta t}{\tau_i} \text{ with } \tau_i = \frac{0.849}{4\,\lambda_h\,\Omega_t}, \quad K_\omega = \frac{\ell_t/R_t}{\lambda_h\,\Omega_t}. \tag{81}$$

The firmware loop period Δt is a firmware constant, and Ω_t is the nominal governed rotor speed—a configuration-time constant taken equal to the governor's setpoint, the same headspeed at which $\bar{\theta}$ was observed. Any unit conversion of $\bar{\theta}$ from degrees to radians happens before (80). The map (80)–(81) is evaluated once, at initialisation time, and $(K_u, K_v, K_\tau, K_\omega)$ are stored; Ω_t is not consulted again on the live control path. Runtime deviations of the actual rotor speed from Ω_t (governor sag, transients) are analysed in Section 8.2.

7.2 Parameters, Inputs, State, Outputs

With the initialisation-time mapping of Section 7.1 settled, what remains is the live-tick specification: what arrives each tick, what carries over between ticks, what is recomputed and discarded, and what is emitted to the servo. In what follows, "state" means a value carried from one control tick to the next; quantities such as $v_{i,\mathrm{ss}}$ and τ_i are recomputed inside each tick and then discarded.

Fixed coefficients. $K_u, K_v, K_\tau, K_\omega$ from Section 7.1. On the live control path the dimensional rotor and air parameters $\rho, A_t, \Omega_t, R_t, \sigma, a, \ell_t$ appear nowhere; the loop period Δt and the apparent-mass coefficient $\kappa = 8/(3\pi) \approx 0.849$ have both been absorbed into K_τ. If the loop rate changes, K_τ must be recomputed.

Inputs (provided each control tick):

$$u(t), \qquad \hat{V}_\infty^{\mathrm{aero}}(t), \qquad \hat{V}_\perp(t), \qquad \omega(t) \quad [\mathrm{rad/s}]. \tag{82}$$

u is the dimensionless thrust command from the yaw outer loop, normalised so that $u = 1$ corresponds to the reference thrust T_{ref} used to define K_u and K_v. It is taken positive in the thrust-producing direction used throughout Section 3. $\hat{V}_\infty^{\mathrm{aero}}$ and $\hat{V}_\perp$ come from the airframe's airspeed estimator, non-dimensionalised by v_{ref} (or set to zero in near-hover applications). The body yaw rate ω is taken from the same gyro signal the outer yaw loop uses; together with K_ω, it gives the axial-flow contribution discussed in Section 3.3. Inside the tick the inverse uses the composite

$$\hat{V}_\infty(t) \;=\; \hat{V}_\infty^{\mathrm{aero}}(t) \;+\; K_\omega\,\omega(t), \tag{83}$$

which is the only place ω enters the algebra below.

State (one scalar, evolved by the inverse block):

$$\hat{v}_i(t). \tag{84}$$

$\hat{v}_i$ is the current dimensionless induced-velocity estimate. It is the only memory in the inverse block.

Algebraic quantities (computed inside each control tick):

$$\hat{v}_{i,\mathrm{ss}}(t). \tag{85}$$

$\hat{v}_{i,\mathrm{ss}}$ is the dimensionless steady inflow target implied by the current command and flight condition; it is recomputed each tick and discarded.

Outputs :

$$\theta_t(t) \quad [\text{rad, to the servo}]. \tag{86}$$

7.3 The Update Law

At each control tick of period Δt, keep the order of ideas straight. First, the command and flight condition define the steady inflow target $\hat{v}_{i,\mathrm{ss}}$. Second, the current state $\hat{v}_i$, not $\hat{v}_{i,\mathrm{ss}}$, is used to compute the pitch needed right now. Third, the discrete inflow-update coefficient K_τ sets how far $\hat{v}_i$ moves toward $\hat{v}_{i,\mathrm{ss}}$ during this tick:

$$\boxed{\begin{aligned} u &= \hat{v}_{i,\mathrm{ss}}\sqrt{(\hat{V}_\infty + \hat{v}_{i,\mathrm{ss}})^2 + \hat{V}_\perp^2}, && (87)\\ \theta_t &= K_u\, u \;+\; K_v\,(\hat{v}_i + \hat{V}_\infty), && (88)\\ \hat{v}_i &\leftarrow \hat{v}_i \;+\; K_\tau\, \hat{v}_i\,(\hat{v}_{i,\mathrm{ss}} - \hat{v}_i). && (89)\end{aligned}}$$

These are the three boxed equations (67), (71), and (75) of Section 6, reordered into causal control-tick order. The only implicit step is (87), a scalar solve for $\hat{v}_{i,\mathrm{ss}}$. In hover it has a closed form ($\hat{v}_{i,\mathrm{ss}} = \sqrt{u}$); with axial or edgewise flow it can be solved with a few Newton iterations because the positive working-state branch is one-dimensional. The pitch equation (88) then uses $\hat{v}_i$, not $\hat{v}_{i,\mathrm{ss}}$, because pitch must cancel the inflow that exists at the blade element now. The update (89) is what moves the estimate toward the future steady value.

Near rotor spin-up or near zero commanded thrust, the state $\hat{v}_i$ itself appears as a factor in (89) (it carries the state-dependent Pitt–Peters time constant), so the effective lag can grow without bound. A practical implementation therefore clamps $\hat{v}_i$ and $\hat{v}_{i,\mathrm{ss}}$ to small physically meaningful lower bounds before applying the update.

7.4 Common Simplifications

Several simplifications are commonly made when fitting the inner loop into a constrained embedded environment.

Near-hover $\hat{V}_\infty, \hat{V}_\perp \approx 0$. The momentum closure (87) collapses on the positive-thrust branch to

$$\hat{v}_{i,\mathrm{ss}} \;=\; \sqrt{u}. \tag{90}$$

This avoids the scalar solve and the airspeed estimate. The simplification is sound for a tail rotor whose airframe spends most of its time in near-hover or low-speed translation; the error grows with edgewise flow and should be re-examined for operating points well outside hover.

Axial-only flow $\hat{V}_\perp = 0$. If in-plane flow is neglected but axial flow through the tail rotor is kept, the momentum closure is no longer a general scalar solve. On the normal working branch, where $\hat{V}_\infty + \hat{v}_{i,\mathrm{ss}} \geq 0$, it reduces to a quadratic:

$$\hat{v}_{i,\mathrm{ss}} = \frac{\sqrt{\hat{V}_\infty^2 + 4u} - \hat{V}_\infty}{2}. \tag{91}$$

This expression reduces to the near-hover result when $\hat{V}_\infty = 0$. It is useful when the implementation has a reasonable axial-flow estimate (typically dominated by the yaw-rate term $K_\omega\,\omega$) but does not want to solve the full edgewise-flow equation each tick. For strongly negative axial flow, the branch condition should be checked and the guarded scalar solve in (87) should be used instead.

Equations (87)–(89), together with the four coefficients $K_u, K_v, K_\tau, K_\omega$ of Section 6, are the deliverable of this note from a control-implementation standpoint. Representative magnitudes for the underlying physical parameters appear in Section 8.

8 Parameter Selection and Sensitivity

Section 7.1 reduced the loop's tunables to three exposed parameters: hover trim pitch $\bar{\theta}$, blade loading parameter σa, and boom-to-tail-radius ratio ℓ_t/R_t. The stored coefficients $(K_u, K_v, K_\tau, K_\omega)$ are derived from them at initialisation time via (80)–(81). This section answers three practical questions about those three numbers: what values are typical across heli sizes (Section 8.1), how sensitive the loop is to error in each (Section 8.2), and which parameter to revisit when the airframe setup changes (Section 8.3). A worked 700-class arithmetic example closes the section (Section 8.4).

8.1 Exposed Parameters Across Heli Classes

The three exposed parameters vary surprisingly little across the class range. σa is fixed almost entirely by tail-blade choice (two-bladed symmetric foil), ℓ_t/R_t by airframe convention, and only $\bar{\theta}$ responds materially to airframe mass and aerodynamic loading. Representative values for three classes spanning an order of magnitude of airframe mass are collected in Table 1, together with the resulting stored coefficients at a $\Delta t = 1\,\text{ms}$ loop period. Geometric inputs are class medians from the airframe survey of Appendix A; $\bar{\theta}$ is a nominal hover trim value typical of mild 3D loading.

Quantity	450 class	550 class	700 class
Main blade [mm]	325	550	700
Flying mass [kg]	0.75	3.0	5.0
R_t [mm]	80	127	140
c [mm]	18	24	30
ℓ_t [mm]	450	780	950
Ω_t [rad/s]	1370	980	1036
$\Omega_t R_t$ [m/s]	110	124	145
Exposed parameters:			
$\bar{\theta}$ [deg]	6.0	7.0	8.6
σa	0.82	0.69	0.78
ℓ_t/R_t	5.6	6.1	6.8
Derived stored coefficients:			
λ_h	0.048	0.051	0.062
K_u [deg]	1.9	2.6	3.3
K_v [deg]	4.1	4.4	5.3
K_τ at $\Delta t = 1\,\text{ms}$	0.31	0.24	0.30
K_ω [s]	0.086	0.122	0.107

Table 1: Representative exposed parameters and derived stored coefficients for three heli classes. K_u and K_v are reported in degrees (the firmware unit of $\bar{\theta}$); by construction $K_u + K_v = \bar{\theta}$ at hover. K_τ is dimensionless; K_ω has units of seconds (it multiplies ω in rad/s).

Across the range $\bar{\theta}$ varies by under 50% and σa by under 20%; the discrete inflow gain K_τ sits near 0.3 throughout—a near-similarity result, since $\lambda_h \Omega_t \propto v_h$ (and hence $1/\tau_i$) is set by tail-rotor disc loading, which is approximately constant across the classes.

8.2 Sensitivity to Parameter Errors

How tolerant is the loop to a wrong value in each stored parameter (the three exposed parameters of Section 7.1 and the governor-sourced rotor speed Ω_t)? Implicit differentiation of the static quadratic (80), using $\partial f/\partial\lambda_h = (2K_u + K_v)/\lambda_h$ and writing fractional sensitivities (elasticities) as ratios of logarithmic differentials, gives the closed-form result

$$\frac{d\ln\lambda_h}{d\ln\bar{\theta}} = \frac{K_u + K_v}{2K_u + K_v}, \qquad \frac{d\ln\lambda_h}{d\ln(\sigma a)} = \frac{K_u}{2K_u + K_v}. \tag{92}$$

The two elasticities sum to $(K_u{+}K_v)/(2K_u{+}K_v){+}K_u/(2K_u{+}K_v) = (2K_u{+}K_v)/(2K_u{+}K_v) = 1$, reflecting that λ_h is fixed by the product $\sigma a\cdot\bar{\theta}$ up to the quadratic correction. At the 700-class operating point of Table 1 ($K_u = 3.3$, $K_v = 5.3$) the elasticities evaluate to $8.6/11.9 = 0.72$ and $3.3/11.9 = 0.28$: a 10% increase in $\bar{\theta}$ raises λ_h by 7.2%, and a 10% increase in σa raises it by 2.8%.

Propagating these through (81) (using $K_u \propto \lambda_h^2/\sigma a$, $K_v \propto \lambda_h$, $K_\tau \propto \lambda_h\Omega_t\Delta t$, and $K_\omega \propto (\ell_t/R_t)/(\lambda_h\Omega_t)$) yields the fractional sensitivities of the stored coefficients:

+10% change in	K_u	K_v	K_τ	K_ω
$\bar{\theta}$	+14%	+7%	+7%	−7%
σa	−4%	+3%	+3%	−3%
ℓ_t/R_t	—	—	—	+10%
Ω_t	—	—	+10%	−10%

What the table says. The first row is dominated by K_u, which is quadratic in λ_h; the second row is muted because the response of λ_h to σa is roughly cancelled by the explicit σa in the denominator of $K_u = 12\lambda_h^2/\sigma a$. The third and fourth rows are exact—ℓ_t/R_t and Ω_t enter only the dynamic coefficients K_ω (and, for Ω_t, also K_τ) and do not perturb K_u or K_v at all. Crucially, the *hover static gain is invariant*: by construction $K_u + K_v = \bar{\theta}$ at $u = 1$, regardless of how σa splits the sum between the two coefficients and independent of ℓ_t/R_t or Ω_t. Errors in the stored parameters therefore appear only off-hover (in the curvature of the command–pitch map set by the $K_u u$ term) and in the dynamics (the inflow lag set by K_τ and the yaw-rate axial coupling set by K_ω).

- $\bar{\theta}$ is the most influential of the three user-tunable parameters, but also the easiest to set correctly: it is observed directly on telemetry during a stationary hover. A 10–20% error is unlikely once a single hover trim has been read.
- σa has weak effect. A coarse default of 0.75–0.80 covers all two-bladed symmetric-foil tail rotors to better than 5% on the stored coefficients, and the static hover relation is unaffected. There is little reason to expose this knob to the user.
- ℓ_t/R_t enters only K_ω, which is multiplied by ω. At small yaw rates the contribution is negligible; at a $500\,^\circ\,\mathrm{s}^{-1}$ 3D pirouette ($\omega \approx 8.7\,\mathrm{rad\,s}^{-1}$) a 10% error in ℓ_t/R_t corresponds to roughly a 0.8 m/s error in the effective axial flow, perturbing the operating point without destabilising the loop.
- Ω_t is the nominal governed rotor speed, taken from the rotor governor's configured setpoint rather than tuned independently. It enters only the dynamic coefficients (K_τ linearly, K_ω inversely) and leaves the hover static gain untouched. The standing headspeed should be re-entered—and $\bar{\theta}$ re-observed—if the governor target changes by more than $\sim 10\%$; runtime deviations of the actual rotor speed from this stored value are analysed in the dedicated paragraph below.

Runtime deviations of the actual rotor speed. The stored $(K_u, K_v, K_\tau, K_\omega)$ freeze Ω_t at its nominal value. The actual rotor speed deviates from this on at least two timescales: brief headspeed sag during high-load segments (autorotation entries, heavy collective pumps, sharp yaw demands) and slower drift in the governor's standing target between flights. Writing $\Omega_t^{\mathrm{act}} = (1+\epsilon)\,\Omega_t$ with ϵ the fractional deviation, three effects appear:

- *Frozen-wake gain.* BEMT thrust at fixed pitch scales as $(\Omega_t R_t)^2$ (eq. (16)), so the actual rotor delivers $(1+\epsilon)^2 \approx 1+2\epsilon$ times the nominal thrust for the pitch command the inverse produces. A 10% sag is therefore a $\sim 20\%$ DC plant-gain shortfall—comparable to the inflow-softening variation across the operating envelope (Section 5.6) and absorbed by the outer yaw loop's integral action.

- *Inflow lag.* The wake-relaxation time constant $\tau_i = 0.849/(4\,\lambda_i\,\Omega_t)$ shortens roughly linearly with Ω_t^{act} (eq. (34)), so the actual τ_i is $(1-\epsilon)$ times its nominal value. The stored K_τ over-discretises by the same fraction, slightly slowing the state's chase of $\hat{v}_{i,\mathrm{ss}}$; the effect is small and of correct sign in the lead–lag analysis.

- *Yaw-rate axial flow.* The dimensional contribution $\ell_t\omega$ does not depend on Ω_t at all, and K_ω was computed once with the nominal Ω_t; the stored $\hat{V}_\infty = K_\omega\,\omega$ therefore mis-states the non-dimensional axial flow at the actual operating point by the fractional change in v_{ref} (itself proportional to Ω_t through $v_h = \lambda_h \Omega_t R_t$), i.e. by ϵ. This is a second-order correction on a first-order term and is negligible against the dominant σa-driven uncertainty.

In short: sustained RPM deviations show up primarily as a DC gain shortfall of order 2ϵ, recoverable only by re-running the initialisation-time map with the new headspeed (or by re-observing $\bar{\theta}$ and reinitialising). Transient sag during a high-load segment is a momentary plant-gain change that the outer yaw loop sees as a short-duration disturbance; no inner-loop scheduling is required at RC envelope amplitudes.

8.3 Re-tuning When the Setup Changes

The two routine triggers for a re-tune are a tail-blade swap and a significant mass change. Both alter $\bar{\theta}$; the first also alters geometry. The remaining triggers (headspeed, boom length, aerofoil) touch only one parameter each, often none.

Different tail blades (length or chord). A change in tail-blade length changes both R_t (swept radius) and ℓ_t/R_t; a change in chord changes $\sigma = 2c/(\pi R_t)$, and hence $\sigma a = 2ca/(\pi R_t)$. Both quantities are geometric and can be read off the new blade specification. Then hover and re-observe $\bar{\theta}$ from telemetry: a smaller disc requires a higher pitch for the same thrust, and vice versa.

Heavier or lighter airframe. Tail thrust at hover scales with main-rotor reactive torque, which in turn scales roughly as $W^{3/2}$ through induced power. A 30% mass increase therefore raises required tail thrust by about 50% and $\bar{\theta}$ by a similar fraction through (80). Re-observe $\bar{\theta}$ after any significant payload change. σa and ℓ_t/R_t are unaffected.

Different headspeed. Ω_t is a configuration-time constant set to the governor's nominal target. Changing the standing headspeed requires updating that constant and re-running the initialisation-time map—λ_h is unchanged (it is derived from $\bar{\theta}$), but $K_\tau \propto \Omega_t$ and $K_\omega \propto 1/\Omega_t$ shift linearly with it. The hover trim pitch $\bar{\theta}$ itself also shifts mildly—thrust at fixed pitch scales as Ω_t^2, so a lower headspeed needs a larger $\bar{\theta}$ to deliver the same tail thrust—and should be re-observed if the headspeed changes by more than $\sim 10\%$. Transient sag from the nominal value is left to the outer loop (Section 8.2).

Longer or shorter tail boom. ℓ_t changes but R_t does not. Update ℓ_t/R_t only. $\bar{\theta}$ shifts mildly in the opposite sense (a longer boom needs less thrust at hover to balance the same main-rotor torque) and should be re-observed.

Different tail aerofoil. Symmetric two-bladed foils all give σa near 0.8 to within $\pm 10\%$. For non-standard cambered foils, recompute $\sigma a = (2c/\pi R_t)\, a$ with the foil's actual lift-curve slope. As the sensitivity table above shows, the loop is forgiving here.

8.4 Worked Example: 700 class

A representative 700-class electric machine (main rotor radius ~ 0.79 m, all-up mass ~ 5 kg) carries the following tail-rotor geometry:

$$\begin{aligned} R_t &= 0.140 \text{ m}, & c &= 0.030 \text{ m}, & N_b &= 2, \\ \Omega_t &= 1036 \text{ rad/s}, & \Omega_t R_t &= 145 \text{ m/s}, & & \\ \sigma &= \tfrac{2\cdot 0.030}{\pi\cdot 0.140} = 0.136, & a &= 5.7, & & \\ \bar{T} &= 12 \text{ N}, & A_t &= 0.0616 \text{ m}^2, & \ell_t &= 0.95 \text{ m}. \end{aligned}$$

Hence

$$\begin{aligned} \bar{v}_i &= \sqrt{12/(2\cdot 1.225\cdot 0.0616)} = 8.92 \text{ m/s}, \\ \tau_i &= 0.849\cdot 0.140/(4\cdot 8.92) \approx 3.3 \text{ ms (Pitt–Peters)}, \\ \sigma a &= 0.136\cdot 5.7 = 0.778, \\ \lambda_h &= \bar{v}_i/(\Omega_t R_t) = 8.92/145 = 0.0615, \\ G_0 &= \tfrac{1.225\cdot 0.0616\cdot 145^2\cdot 0.778}{6} = 206 \text{ N/rad}, \\ k_v &= 1/(4\cdot 1.225\cdot 0.0616\cdot 8.92) = 0.372 \text{ m/(s N)}, \\ \beta &= \frac{\sigma a}{16\lambda_h} = \frac{0.778}{0.984} = 0.79. \end{aligned}$$

The DC softening factor is $S = 1/(1+\beta) = 0.56$: the steady-state thrust delivered for a given blade pitch is about 56% of the open-loop (frozen-wake) value.

Yaw-rate contribution. At a sustained 3D yaw rate of $\omega = 500\,^\circ\,\mathrm{s}^{-1} = 8.73$ rad/s, the axial-flow contribution from yawing alone is

$$\ell_t\omega \;=\; 0.95\cdot 8.73 \;=\; 8.29 \text{ m/s},$$

comparable to the hover induced velocity $\bar{v}_i = 8.92$ m/s itself, and giving $\mu_z = \ell_t\omega/(\Omega_t R_t) = 0.057$ against $\lambda_h = 0.0615$. In a with-thrust pirouette this axial flow reduces the rotor's own $v_{i,\mathrm{ss}}$ substantially; in an against-thrust pirouette of the same magnitude the operating point approaches the windmill-brake state and the smooth-wake momentum closure should be treated with caution.

9 Airframe Yaw Dynamics

The preceding sections derived a map from commanded thrust to blade pitch and packaged it as an inner control loop. This section complements that derivation with the yaw equation of motion for the airframe under a prescribed tail-rotor thrust T_t. Four physical effects enter the balance: the aerodynamic anti-torque of the main rotor, the moment of the tail-rotor thrust about the airframe centre of mass, the moment of inertia of the airframe, and the polar moment of inertia of the main rotor. The first three combine directly through Newton's second law for rotation; the fourth couples in through the main motor (with the help of the governor) and is the subject of Section 9.5.

9.1 Assumptions and Sign Conventions

The derivation rests on the following assumptions, applied throughout the section unless explicitly relaxed.

1. *Pure yaw motion.* The airframe rotates about a single axis, the vertical axis $\hat{z}$ through its centre of mass; pitch, roll, and translational dynamics are decoupled and not considered here.
2. *Two rigid bodies coupled by a shaft.* The system is decomposed into the *airframe* (everything mechanically bolted to the chassis, including the motor stator and tail assembly), with moment of inertia I_z about $\hat{z}$, and the *main rotor* (rotating shaft, hub, and blades), with polar moment of inertia I_M about the same axis. The tail rotor's spin angular momentum is perpendicular to $\hat{z}$ and does not contribute to first order; its small yaw-axis inertia is absorbed into I_z.
3. *Constant shaft-relative main-rotor speed.* The main motor, commanded by the governor, holds the shaft-relative angular speed Ω_M (as measured by the motor commutation feedback) at its set point; $\dot{\Omega}_M = 0$. The implications for the *absolute* rotor speed are developed in Section 9.5.
4. *Lumped main-rotor anti-torque.* The aerodynamic torque that the rotor exerts on its shaft is treated as a single scalar $Q_M \geq 0$ (positive by convention; sign on the rotor is $-Q_M\hat{z}$ when the rotor spins along $+\hat{z}$); its functional dependence on collective pitch, headspeed, and disc loading is not resolved here.
5. *No body aerodynamics.* Fuselage and tail-boom drag moments about $\hat{z}$ are neglected at the yaw rates considered.

The sign convention for the yaw rate is fixed by consistency with the axial-flow convention of Section 2.2: $\omega > 0$ denotes airframe rotation in the same sense as the tail-rotor thrust pushes the airframe. Under this convention the main rotor spins in the *opposite* sense, so that a positive tail thrust counters the main-rotor anti-torque reaction at trim.

9.2 Equations of Motion of the Two Bodies

Newton's second law for rotation about $\hat{z}$, applied separately to the airframe and to the main rotor, yields two scalar equations. For the airframe,

$$I_z\,\dot{\omega} \;=\; \tau_{\text{tail}} \;+\; \tau_{\text{s}\to\text{f}}, \tag{93}$$

where τ_{tail} is the moment of the tail-rotor thrust about the centre of mass and $\tau_{\text{s}\to\text{f}}$ is the torque transmitted to the airframe through the rotor shaft (motor mount and main bearings). For the main rotor, with absolute angular velocity Ω_M^{abs} about $\hat{z}$ (to be identified in Section 9.5),

$$I_M\,\dot{\Omega}_M^{\text{abs}} \;=\; \tau_m \;-\; Q_M, \tag{94}$$

where τ_m is the torque applied by the motor (and gearing, if any) to the rotor, and $-Q_M$ is the aerodynamic anti-torque. Newton's third law relates the shaft transmission to the motor torque,

$$\tau_{\mathrm{s}\to\mathrm{f}} \;=\; -\,\tau_m, \tag{95}$$

the airframe feeling the motor's reaction. Equations (93)–(95) are the complete dynamical content of the section; what remains is to evaluate τ_{tail}, Ω_M^{abs}, and τ_m in turn.

9.3 Moment of the Tail-Rotor Thrust

The tail rotor acts at distance ℓ_t behind the centre of mass (the moment arm already introduced in Section 3.3). Its thrust T_t is directed perpendicular to the tail boom, in the disc plane, so its moment about $\hat{z}$ is, with the sign convention of Section 9.1,

$$\boxed{\tau_{\mathrm{tail}} \;=\; T_t\,\ell_t.} \tag{96}$$

This is the only torque on the yaw axis that is directly commanded: every other contribution in the balance is either a trim load fixed by the main rotor or a kinematic reaction.

9.4 Yaw Equation under a Quasi-Steady Rotor

Consider first the simplification in which both the rotor and the airframe are in steady rotation, $\dot{\Omega}_M^{\mathrm{abs}} = 0$ and $\dot{\omega} = 0$. The rotor equation (94) then reduces to the torque balance $\tau_m = Q_M$, and by (95) the reaction onto the airframe is $\tau_{\mathrm{s}\to\mathrm{f}} = -Q_M$. Substituting into (93) together with (96) gives the elementary yaw equation

$$I_z\,\dot{\omega} \;=\; T_t\,\ell_t \;-\; Q_M. \tag{97}$$

At trim ($\dot{\omega} = 0$) this fixes the hover tail thrust at

$$\bar{T} \;=\; \frac{Q_M}{\ell_t}, \tag{98}$$

the value used throughout Sections 4–7 as the trim operating point of the inner loop. Writing $T_t = \bar{T} + \delta T_t$, the perturbation dynamics about trim are

$$I_z\,\dot{\omega} \;=\; \ell_t\,\delta T_t, \tag{99}$$

a single integrator from the inner-loop thrust over-pressure δT_t to yaw rate ω (and one further integration to heading, if needed).

Equation (99) is the correct yaw equation only insofar as $\tau_m = Q_M$ is exact during transients. It is not: when the airframe yaw-accelerates, the rotor must yaw-accelerate with it, and τ_m must exceed Q_M by whatever inertial term is needed to enforce that motion. The next subsection makes the correction explicit.

9.5 Inertial Coupling Through the Motor

The main-rotor shaft is mounted to the airframe and rotates with it about $\hat{z}$. The rotor's angular velocity in the inertial frame is therefore the sum of its shaft-relative spin and the airframe's yaw rate,

$$\boxed{\Omega_M^{\mathrm{abs}} \;=\; \Omega_M \;+\; \omega.} \tag{100}$$

The motor, with the help of the governor (Section 9.1, item 3), holds Ω_M constant, not Ω_M^{abs}; the absolute rotor speed therefore tracks the airframe yaw rate. Differentiating (100) under $\dot{\Omega}_M = 0$,

$$\dot{\Omega}_M^{\mathrm{abs}} \;=\; \dot{\omega}. \tag{101}$$

The rotor equation (94) then becomes

$$\tau_m \;=\; Q_M \;+\; I_M\,\dot{\omega}, \tag{102}$$

exposing the inertial component $I_M\dot{\omega}$ that the motor must supply on top of the steady anti-torque Q_M whenever the airframe yaw-accelerates. The governor's role is to command this additional torque from the motor as soon as the yaw rate begins to change; the work itself is done by the motor's electrical input. By (95) the same component reacts onto the airframe through the motor mount with opposite sign,

$$\tau_{\text{s}\to\text{f}} \;=\; -\big(Q_M + I_M\,\dot{\omega}\big). \tag{103}$$

Substituting into the airframe equation (93) with (96) and collecting the $\dot{\omega}$ terms on the left,

$$\boxed{(I_z + I_M)\,\dot{\omega} \;=\; T_t\,\ell_t \;-\; Q_M.} \tag{104}$$

The effective yaw inertia is

$$I_z^{\text{eff}} \;\equiv\; I_z + I_M, \tag{105}$$

the sum of the airframe inertia and the rotor's polar inertia. Equivalently, so long as the motor (with the help of the governor) holds Ω_M constant, the airframe and main rotor are rotationally rigid about $\hat{z}$: they share the same $\dot{\omega}$, and the tail rotor accelerates both bodies as a single rigid system. The same conclusion follows from angular-momentum bookkeeping for the combined system: the total angular momentum about $\hat{z}$ is $H_z = I_z\omega + I_M(\Omega_M + \omega)$, and its rate of change under the external torque $\tau_{\text{tail}} + (-Q_M)\hat{z}$ (the only torques external to the combined system, the motor torque τ_m being internal between the rotor and the airframe) reproduces (104) directly.

9.6 Aerodynamic Correction to the Anti-Torque

Equation (104) carries Q_M as a constant external load, but Q_M is in fact set by the blade tip speed relative to the air. At radius r_M the tip speed is $\Omega_M^{\text{abs}} r_M = (\Omega_M + \omega)\,r_M$, so the elemental dynamic pressure scales as $(\Omega_M + \omega)^2$ and, at fixed blade pitch, so does the integrated anti-torque. Expanding to first order in $\omega/\Omega_M \ll 1$,

$$Q_M(\omega) \;\approx\; Q_M(0)\,\Big(1 \;+\; 2\,\omega/\Omega_M\Big). \tag{106}$$

At $\omega = 500\,^\circ\,\text{s}^{-1} \approx 8.7\,\text{rad/s}$ and a 700-class main-rotor speed $\Omega_M \approx 220\,\text{rad/s}$, the correction is $2\omega/\Omega_M \approx 8\%$. This is a slow disturbance on the trim, absorbed by the outer loop, and is not carried as an inner-loop feedforward term. The analogous tail-rotor effect—axial inflow $\ell_t\omega$ of order the hover induced velocity $\bar{v}_i$, see Section 3.3—is two orders larger in relative magnitude and is the reason the inner inverse retains the explicit yaw-rate term $K_\omega\,\omega$ in (83).

9.7 Summary and Worked Example

Combining (104) and (98) and writing $T_t = \bar{T} + \delta T_t$ gives the control-oriented yaw equation

$$\boxed{(I_z + I_M)\,\dot{\omega} \;=\; \ell_t\,\delta T_t,} \tag{107}$$

a single integrator from inner-loop thrust over-pressure to yaw rate, with open-loop gain $\ell_t/(I_z + I_M)$. Three remarks collect the structural content of the derivation.

(i) Trim. At trim the tail rotor produces $\bar{T} = Q_M/\ell_t$; the entire inner-loop derivation is built on this operating point.

(ii) Effective inertia. The relevant inertia for yaw transients is $I_z + I_M$, not I_z alone, because the motor (with the help of the governor) enforces shaft-relative speed and therefore locks the main rotor's yaw acceleration to the airframe's. Sizing the outer loop's required tail authority from the bare airframe inertia would underestimate the time required to reach a given yaw rate by a factor $(I_z + I_M)/I_z$.

(iii) Governor bandwidth. Equation (104) is exact in the infinite-bandwidth governor limit. A finite-bandwidth governor admits $\dot{\Omega}_M \neq 0$ during transients, adding a term $I_M \dot{\Omega}_M$ to the right-hand side of (104). The effect manifests as a transient disturbance on the yaw axis and a plant-gain shift on the inner loop; the latter is the same effect catalogued under "governor sag" in Section 8.2.

Worked example (700 class). Extending the worked example of Section 8.4 ($m = 5\,\mathrm{kg}$, $\ell_t = 0.95\,\mathrm{m}$, main-rotor radius $R_M = 0.79\,\mathrm{m}$, $\Omega_M \approx 220\,\mathrm{rad/s}$, $\bar{T} = 12\,\mathrm{N}$), the inertias are estimated as

$$
\begin{aligned}
I_z &\approx m\,r_g^2 \approx 5 \cdot 0.40^2 = 0.80\ \mathrm{kg\,m^2} \qquad (r_g \approx 0.40\ \mathrm{m},\ \text{airframe radius of gyration}),\\
I_M &\approx \tfrac{1}{3} N_{b,M}\, m_b\, R_M^2 \approx \tfrac{1}{3} \cdot 2 \cdot 0.13 \cdot 0.79^2 = 0.05\ \mathrm{kg\,m^2} \qquad (\text{thin-rod approximation, two } 130\,\mathrm{g} \text{ main blades})
\end{aligned}
$$

The main rotor contributes about 6% of the effective inertia, $I_z + I_M \approx 0.85\,\mathrm{kg\,m^2}$, and the steady anti-torque implied by the trim thrust is $Q_M = \bar{T}\,\ell_t = 12 \cdot 0.95 = 11.4\,\mathrm{N\,m}$. The open-loop yaw gain evaluates to

$$
\frac{\ell_t}{I_z + I_M} = \frac{0.95}{0.85} \approx 1.12\ \mathrm{rad\,s^{-2}}\ \text{per N},
$$

so a $1\,\mathrm{N}$ inner-loop over-pressure produces about $64^\circ\,\mathrm{s^{-2}}$ of yaw acceleration. At $\omega = 500\,^\circ\,\mathrm{s^{-1}}$ the aerodynamic correction (106) amounts to $\Delta Q_M \approx 0.9\,\mathrm{N\,m}$, equivalent to a $\Delta\bar{T} \approx 1\,\mathrm{N}$ drift in the inner-loop trim, well within the outer-loop disturbance budget.

10 Caveats

The model developed in Sections 2–7 rests on a chain of approximations: thin-foil aerodynamics, momentum-theory inflow, a first-mode Pitt–Peters lag, a perfectly governed tip speed, and an ideal tail servo. Each is good enough to derive a working inverse, but each has known limits that should be flagged before the coefficients of Section 7.1 are taken to the bench. The points below are ordered roughly from those that most directly perturb the stored coefficients to those that act outside the inverse block.

Inflow time constant. Pitt–Peters time constants are systematically low compared to flight-test data on full-scale rotors (factor of 2–4 underestimate); RC practitioners often use $\tau_i \approx 0.1$–0.3 rotor revolutions empirically. For design, the Pitt–Peters value should therefore be treated as a lower bound on the inflow lag, and either inflated by a fitted multiplier or verified against bench data before being used to set compensator coefficients. The coefficient K_τ in (81) inherits this same uncertainty and is the natural place to absorb a fitted correction.

Low-Reynolds aerodynamics. Blade lift-curve slope drops noticeably at the low Reynolds numbers ($Re \sim 30\,000$–$80\,000$) typical of RC tail-rotor tips, especially with thin symmetric sections, and section drag rises correspondingly. The default $\sigma a \approx 0.8$ of Section 7.1 assumes near-2D thin-airfoil behaviour; for very small rotors or unusual sections the exposed σa should be scaled down accordingly.

Blade planform. Solidity calculations as written assume rectangular blades. Tapered or paddle blades need an equivalent solidity weighted by $r^2 c(r)$, which concentrates the lift contribution toward the tip and slightly raises the effective σa relative to a chord-averaged estimate.

Main-rotor wake. The tail rotor sits inside the main-rotor wake, so $V_\perp$ in (25) is dominated by main-rotor downwash—the main source of tail-rotor effectiveness loss in turns—and not by airframe sideslip alone. A simple aero airspeed estimator does not see this contribution. The pragmatic options are to augment $\hat{V}_\perp$ with a main-rotor thrust-dependent term, or to set $\hat{V}_\perp \equiv 0$ (the near-hover simplification of Section 7.4) and absorb the lost authority into the outer loop's headroom.

Yaw-rate axial flow. The axial flow V_∞ is dominated in 3D flight by the yaw-rate contribution $\ell_t \omega$ rather than by the airframe-translation component V_∞^{aero} (Section 3.3); in slow forward flight or sustained hover the two contributions can be of comparable magnitude. The $K_\omega\,\omega$ term (83) captures the dominant component for free; the airspeed-based correction is optional and only earns its keep on airframes that fly meaningful translational speeds.

Governor and tail servo. Two unmodelled blocks bracket the inverse derived here. The upstream governor holds Ω_t constant only within its own loop bandwidth; the inverse takes Ω_t as a fixed parameter at the governor's nominal target, so transient sag from that target shows up as a mismatch between the stored coefficients and the actual plant (Section 8.2). Downstream, the commanded θ_t reaches the blade root only through a servo and linkage with their own bandwidth, deadband, and possible backlash. The inverse is exact in the limit that both blocks are ideal; deviations show up as a plant-gain shift at the input of the inner loop and as a static error at its output, respectively, and are best characterised empirically on the airframe rather than modelled from datasheets.

A Example Helicopters

This appendix lists nominal tail-rotor parameters for ten contemporary RC helicopters spanning the 450–700 class range. Values are drawn from manufacturer-published airframe and blade specifications and from representative 3D operating headspeeds. The tail radius R_t is the swept radius (hub plus blade), the chord c is the nominal tail-blade chord supplied with the kit, and the moment arm ℓ_t is the geometric distance from the main shaft to the tail hub, estimated from overall airframe length and main-shaft position where it is not directly published. The tail rotor angular speed Ω_t is computed from a representative 3D main-rotor headspeed and the published main:tail gear ratio for each model. All listed tail rotors are two-bladed ($N_b = 2$), so solidity reduces to $\sigma = 2c/(\pi R_t)$; tip speed $\Omega_t R_t$ is derived. Flying mass is the manufacturer-stated all-up weight including battery and typical electronics.

A.1 450 class — 325 mm blades

Quantity	T-Rex 450 Pro	Blade 450	Gaui X3
Main blade [mm]	325	325	325
Tail blade [mm]	62	58	62
Flying mass [kg]	0.75	0.75	0.75
R_t [mm]	80	71	82
c [mm]	18	17	17
ℓ_t [mm]	460	440	450
Headspeed [RPM]	3000	2800	3000
Gear ratio (m:t)	4.08	4.66	4.69
Tail RPM	12240	13050	14070
Ω_t [rad/s]	1281	1366	1473
$\Omega_t R_t$ [m/s]	102	97	121
σ	0.143	0.152	0.132

A.2 420 class

Quantity	OMP M4	OMP M4 Max	Goosky RS4	SAB Goblin 420	Steam AK420
Main blade [mm]	385	420	390	420	420
Tail blade [mm]	71	71	68	70	70
Flying mass [kg]	1.35	1.50	1.30	1.70	1.20
R_t [mm]	90	95	93	95	95
c [mm]	20	20	20	20	20
ℓ_t [mm]	550	560	550	580	550
Headspeed [RPM]	2850	2800	2900	3000	2800
Gear ratio (m:t)	4.70	4.70	4.70	4.00	4.05
Tail RPM	13400	13160	13630	12000	11340
Ω_t [rad/s]	1400	1378	1430	1257	1188
$\Omega_t R_t$ [m/s]	126	131	133	119	113
σ	0.141	0.134	0.137	0.134	0.134

A.3 500 class

Quantity	T-Rex 500 Pro	T-Rex 500X	Blade 500	Goblin 500
Main blade [mm]	425	470	425	500
Tail blade [mm]	70	78	70	80
Flying mass [kg]	1.7	2.0	1.8	2.4
R_t [mm]	103	110	99	113
c [mm]	22	22	20	22
ℓ_t [mm]	580	620	600	680
Headspeed [RPM]	2700	2700	2500	2500
Gear ratio (m:t)	4.03	3.81	4.66	4.00
Tail RPM	10880	10290	11650	10000
Ω_t [rad/s]	1140	1080	1220	1047
$\Omega_t R_t$ [m/s]	117	119	121	118
σ	0.136	0.127	0.129	0.124

A.4 550 class

Quantity	T-Rex 550X	OMP M5	Goosky RS5	Goblin 570	Logo 550
Main blade [mm]	550	550	550	570	550
Tail blade [mm]	95	95	95	95	100
Flying mass [kg]	3.3	2.3	3.5	3.2	3.0
R_t [mm]	127	125	127	130	135
c [mm]	24	24	24	24	24
ℓ_t [mm]	780	780	780	800	800
Headspeed [RPM]	2400	2400	2400	2300	2200
Gear ratio (m:t)	3.85	3.80	3.86	4.50	4.50
Tail RPM	9240	9120	9260	10350	9900
Ω_t [rad/s]	968	955	970	1084	1037
$\Omega_t R_t$ [m/s]	123	119	123	141	140
σ	0.120	0.122	0.120	0.118	0.113

A.5 700 class

Quantity	T-Rex 700	OMP M7	Goosky RS7	Goblin 700
Main blade [mm]	700	715	700	690
Tail blade [mm]	105	106	105	105
Flying mass [kg]	5.0	5.3	5.0	5.0
R_t [mm]	140	146	140	142
c [mm]	30	30	30	30
ℓ_t [mm]	950	970	970	960
Headspeed [RPM]	2125	2050	2050	2050
Gear ratio (m:t)	4.66	4.70	4.70	4.90
Tail RPM	9900	9640	9640	10050
Ω_t [rad/s]	1036	1010	1009	1052
$\Omega_t R_t$ [m/s]	145	147	141	149
σ	0.136	0.131	0.136	0.134